

Buck变换器技术概述

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Agenda:

- 一、Buck 拓扑及功率器件选型介绍
- 二、Buck 变换器常见控制模式介绍
- 三、环路分析及补偿设计(PCM)
- 四、EMI 与 PCB 布板设计
- 五、Buck 常见问题及解决方法

Agenda:

一、**Buck** 拓扑及功率器件选型介绍

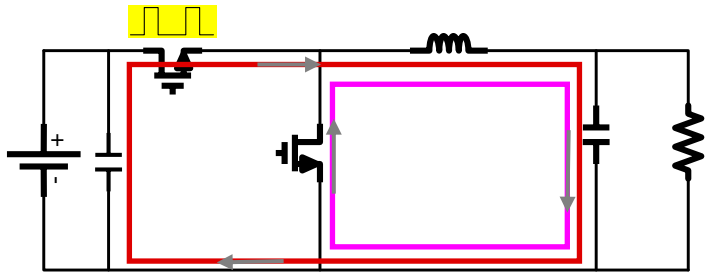
二、Buck 变换器常见控制模式介绍

三、环路分析及补偿设计(PCM)

四、EMI 与 PCB 布板设计

五、Buck常见问题及解决方法

Buck Circuit Operating Principle:



According to Volt-second balance principle:

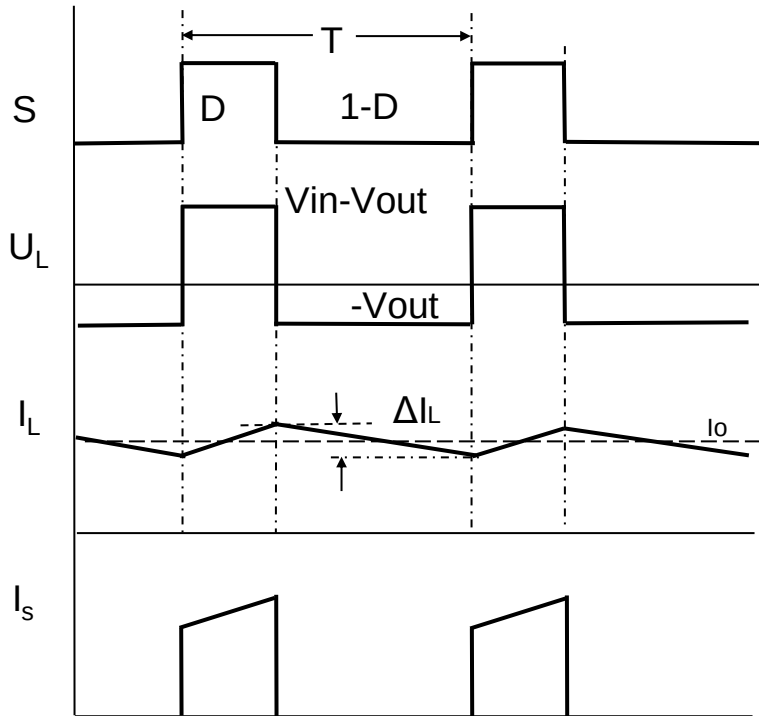
$$(V_{in} - V_{out})DT_s = V_{out}(1 - D)T_s$$

$$\Rightarrow V_{out} = DV_{in}$$

According to the basic equation of L in 1-D time:

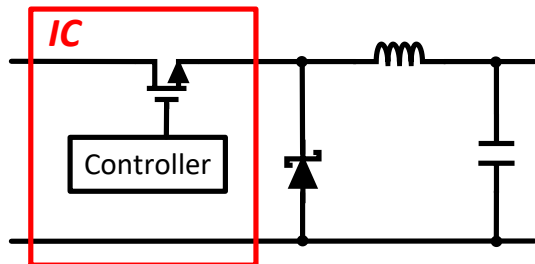
$$\Delta I_L = (V_{in} - V_{out})DT_s / L$$

$$L = \frac{V_{out}(V_{in} - V_{out})}{V_{in}kI_{out}f_{sw}}$$



Synchronous and non-synchronous buck

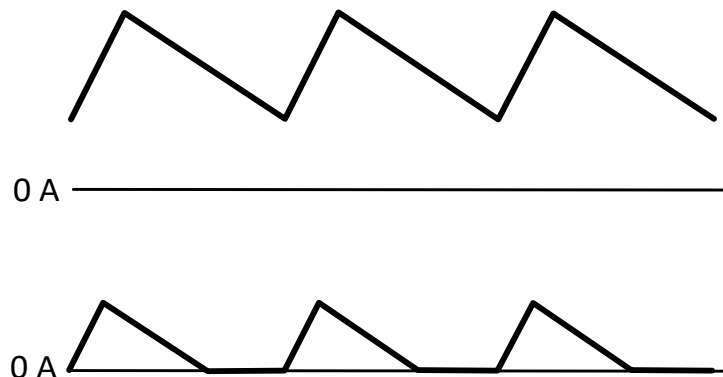
Non-Synchronous Buck



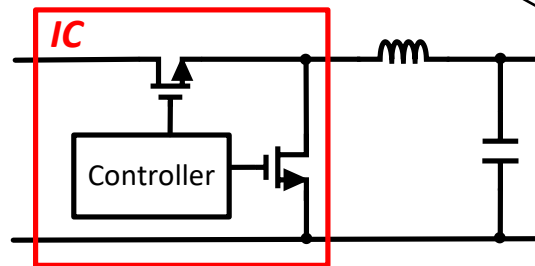
Heavy Load

Light Load

Non-sync

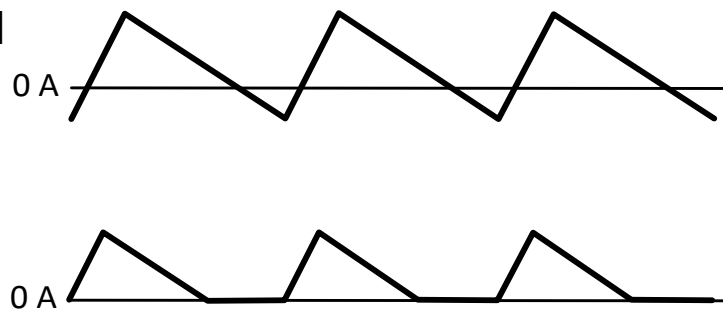


Synchronous Buck



Sync-FCCM

Sync-DCM
(Diode
Emulation)



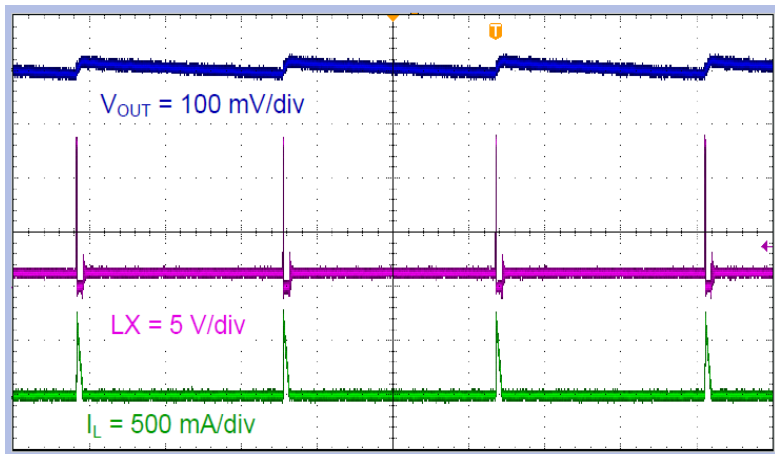
Synchronous and non-synchronous buck

	Non-Synchronous	Synchronous
Size / Ease of use	Larger size	Smaller size, Easier to Use
Light Load Efficiency	Better	Worse with FCCM, better with DCM
Heavy Load Efficiency	More conduction loss Less switching loss	Less conduction loss More switching loss due to reverse recovery Adding small Schottky gives highest efficiency
Lower V_{OUT}	Lower Efficiency (V_F/V_{OUT})	Higher Efficiency
Thermal	Better with two packages	Harder with one package Adding small Schottky reduces power loss a lot
Fixed F_{SW}	No	Yes with FPWM, No with DCM
Noise	less	More, due to body diode reverse recover charge Adding small Schottky diode reduces noise a lot
Unloading Transient	Worse	Good with FPWM, worse with DCM
Controllability	No LS control	More control flexibility

Light Load Condition Waveform



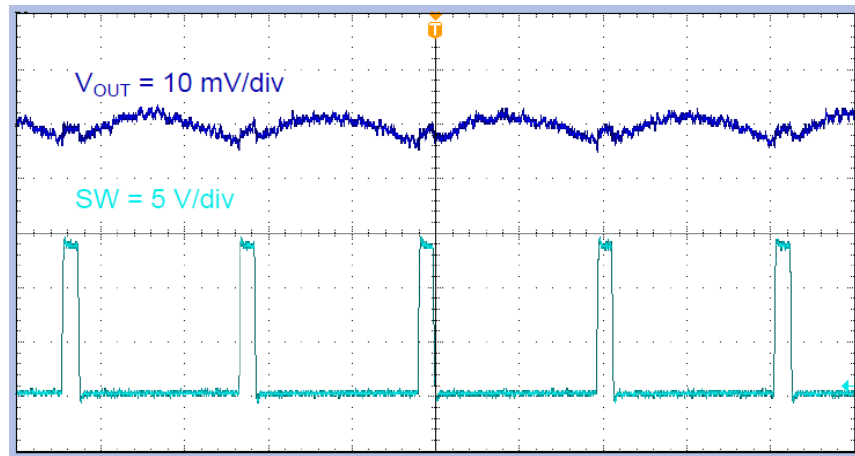
Advanced Eco-Mode
(TPS56X201)



(20 $\mu\text{s/div}$, $I_{out} = 0.01\text{A}$)



Force PWM Mode
(TPS56X208)



(800ns/div, $I_{out} = 0.01\text{A}$)

Input Capacitors Selection

Design requirements (TPS54335A):

DESIGN PARAMETER	EXAMPLE VALUE
Input voltage range	8 to 28 V
Output voltage	5 V
Transient response, 1.5-A load step	$\Delta V_O = \pm 5\%$
Input ripple voltage	400 mV
Output ripple voltage	30 mV
Output current rating	3 A
Operating Frequency	340 kHz

Calculating the input ripple voltage (ΔV_{IN}):

$$\Delta V_{IN} = \frac{I_{OUT(MAX)} \times 0.25}{C_{BULK} \times f_{SW}} + (I_{OUT(MAX)} \times ESR_{MAX})$$

For this design, a 10- μ F, X7R dielectric capacitor rated for 35 V is used for the input decoupling capacitor. The ESR is approximately 2 m Ω , and the current rating is 3 A. Additionally, a small 0.1- μ F capacitor is included for high frequency filtering.

Part Number	$V_{IN,Max}$ (V)	V_O (V)	I_O (A)	F_{SW} (kHz)	C_{IN} (μ F)	C_{IN} Chosen (μ F)
TPS54335A	28	5	3	340	5.59	10

Inductor Value Selection

- Inductance equation for traditional Buck Converter (TPS54335A):

$$L_{\text{MIN}} = \frac{V_{\text{OUT}} \times (V_{\text{IN(MAX)}} - V_{\text{OUT}})}{V_{\text{IN(MAX)}} \times K_{\text{IND}} \times I_{\text{OUT}} \times f_{\text{SW}}}$$

– $K = \Delta I_L / I_O$, where I_O is full load current.

– K is usually between 0.1 and 0.4

- The RMS inductor current ($I_L(\text{RMS})$):

$$I_{L(\text{RMS})} = \sqrt{I_{\text{OUT(MAX)}}^2 + \frac{1}{12} \times \left(\frac{V_{\text{OUT}} \times (V_{\text{IN(MAX)}} - V_{\text{OUT}})}{V_{\text{IN(MAX)}} \times L_{\text{OUT}} \times f_{\text{SW}} \times 0.8} \right)^2}$$

- The peak inductor current ($I_L(\text{PK})$):

$$I_{L(\text{PK})} = I_{\text{OUT(MAX)}} + \frac{V_{\text{OUT}} \times (V_{\text{IN(MAX)}} - V_{\text{OUT}})}{1.6 \times V_{\text{IN(MAX)}} \times L_{\text{OUT}} \times f_{\text{SW}}}$$

Part Number	V _{IN,Max} (V)	V _O (V)	I _O (A)	F _{SW} (kHz)	K	L (μH)	L Chosen (μH)
TPS54335A	28	5	3	340	0.3	13.4	15

常用电感的种类和特性

- 蔽磁露:

工字型磁芯绕漆泡线，外面封锰锌类罩子。

高频特性好，屏蔽性好，饱和特性差，价格偏高。



- 磁封胶:

工字型磁芯绕漆泡线，外面封胶(胶水里会有一些粉料)。

高频特性好，饱和特性差，价格便宜。



- 一体成型:

碳基合粉压制而成。

饱和特性好。



电感选取注意事项

- 饱和电流
- 温升电流
- 电感DCR

Electrical Characteristics

Part Number	Inductance (μH)	Tolerance ($\pm\%$)	Test Frequency (MHz)	RDC($\text{m}\Omega$) Max(Typ.)	饱和电流	温升电流
					Isat(A) Max(Typ.)	Irms(A) Max(Typ.)
UHEI201610A-R47M-Q9	0.47	20	2	36(30)	3.5(3.9)	3.1(3.5)
UHEI201610A-1R0M-Q9	1.0	20	2	60(50)	3.0(3.2)	2.7(3.0)

Note: When ordering, please specify tolerance code. Tolerance: M= $\pm 20\%$

- Operating temperature range - 40°C ~ 125°C (Including self - temperature rise)

Isat for Inductance drop 30% from its value with current

Irms for a 40°C temperature rise from 25°C ambient with current



越小
越好

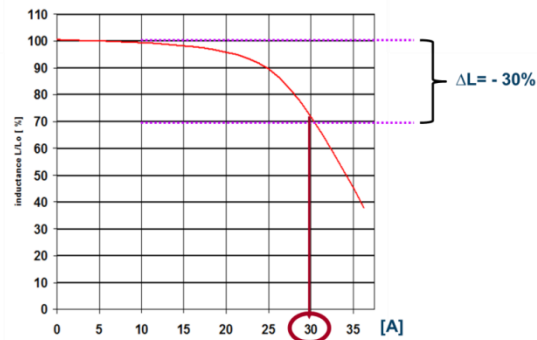


越大
越好



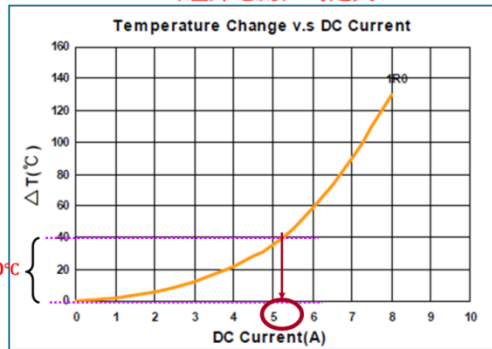
越大
越好

Isat (饱和电流) 的定义



► Isat for Inductance drop 30% from its value without current

Irms (温升电流) 的定义



► Irms for a 40°C temperature rise from 25°C ambient with current

Output Capacitors Selection – TPS54335A

- Load Transient:

$$C_O \geq \frac{2\Delta I_{out}}{f_{sw}\Delta V_{out}} = 35.3\mu\text{F}$$

- The minimum output capacitance required to meet the output voltage ripple specification

$$C_O \geq \frac{I_{ripple}}{8f_{sw}V_{out_ripple}} = 11\mu\text{F}$$

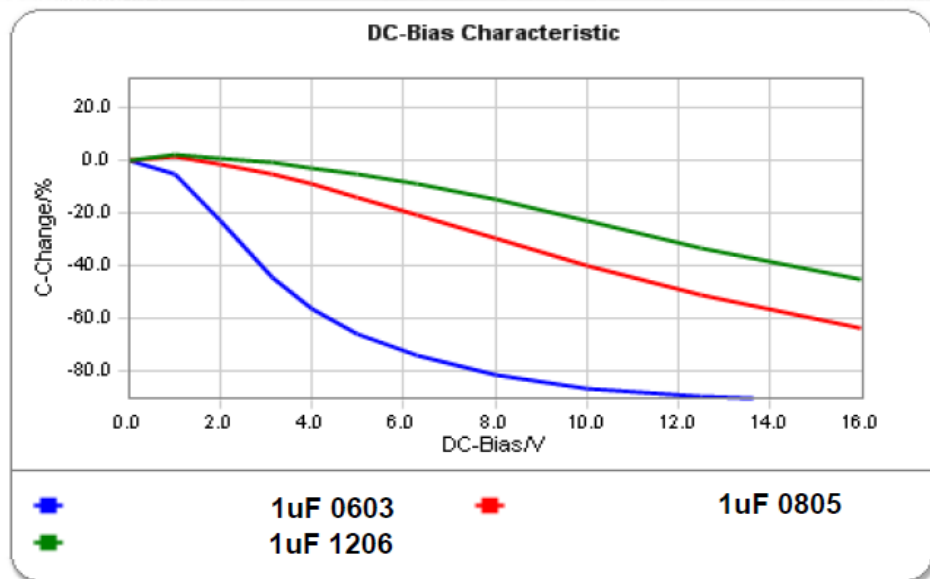
- Select the largest value: 47 μF

Part Number	V _{IN,Max} (V)	V _O (V)	I _O (A)	F _{SW} (kHz)	C _{OUT} (μF)	C _{OUT} Chosen (μF)
TPS54335A	28	5	3	340	35.3	47

电容的电压特性

Voltage Bias Effect Including Case Size

X5R, 16V Rated Capacitors



Voltage de-rating

Class 1 Dielectrics
COG etc do not
require voltage
derating.

Class 2 Dielectrics
such as X7R and
X5R, lose significant
capacitance as you
approach the rated
voltage.

Capacitance decreases more quickly with smaller case sizes

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Control Architecture

Basically, Buck Converters contain 5 blocks :

1. Power Stage/Converter

Power delivery

2. Sensor

Monitor output voltage

3. Controller/Compensator

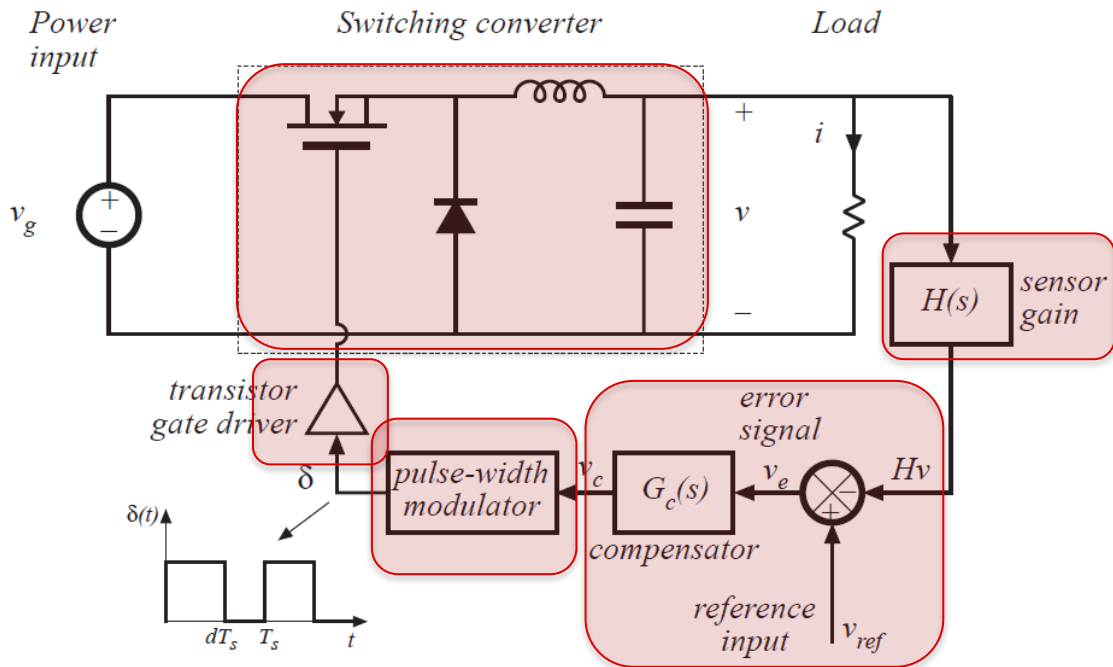
Compare output voltage with reference and compensate the loop gain and phase for nice stability and load transient performance.

4. Modulator

Convert the output of compensator to high frequency MOSFET on/off information.

5. Driver

MOSFET gate driver; dead-time.



Common Control Architecture Summary

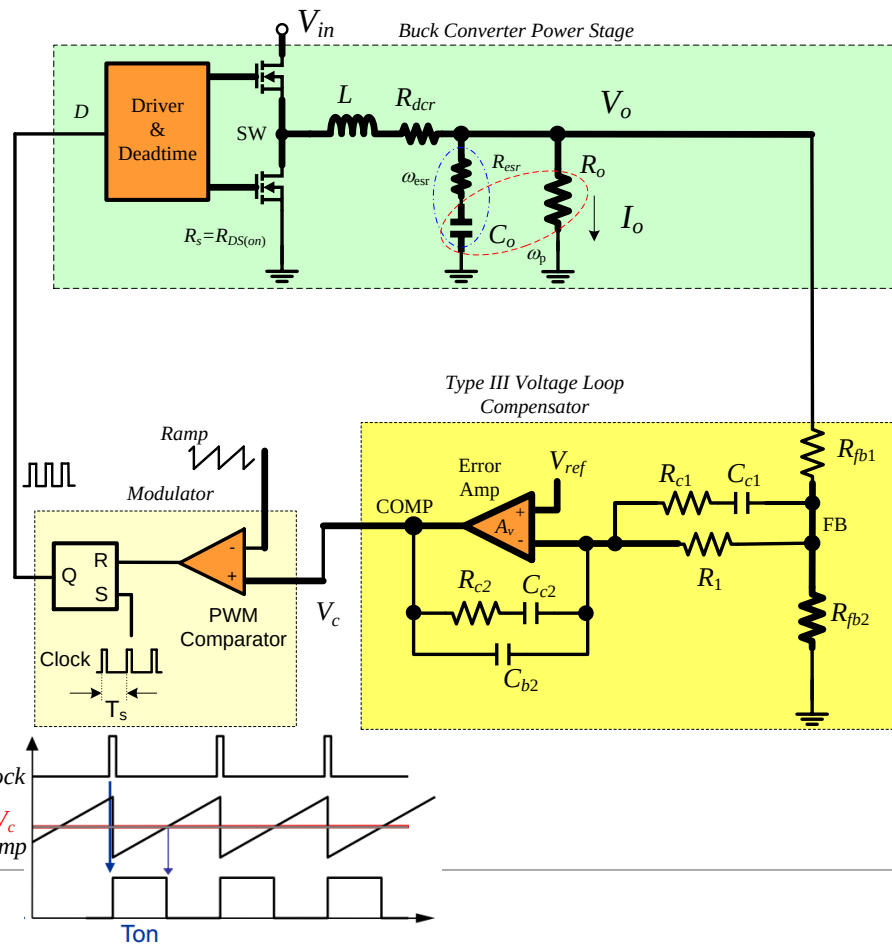
1. Linear Control Architectures

- Voltage Mode
- Peak Current Mode
- ...

2. Non-linear Control Architectures

- Hysteretic Mode
- DCAP Mode
- DCAP2 Mode
- DCAP3 Mode
- ...

Voltage Mode Control (VM)



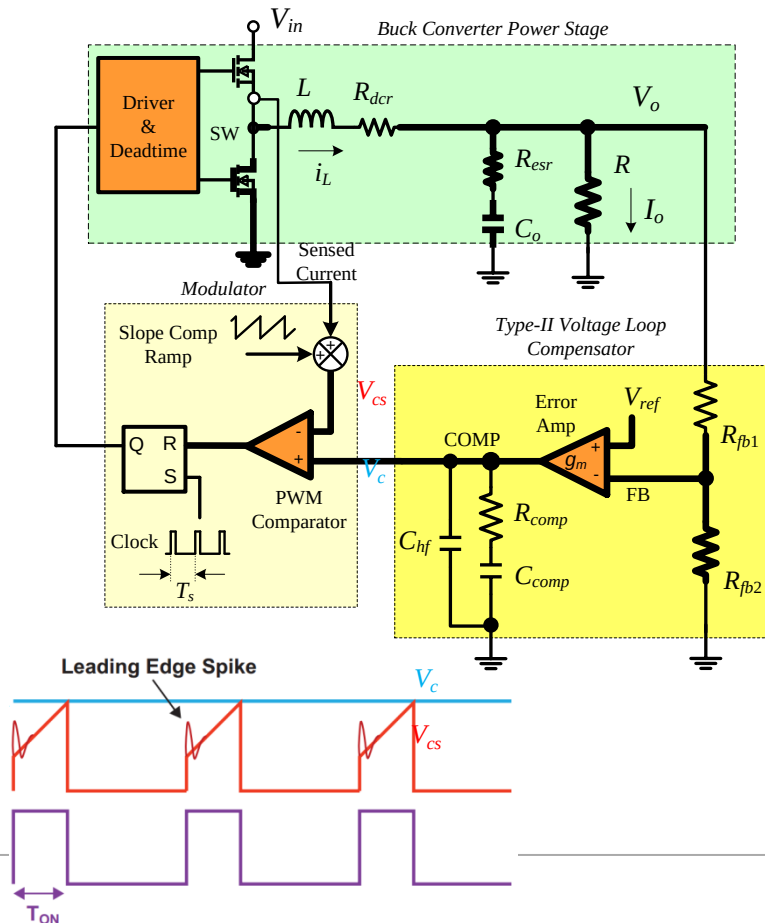
Advantage

- Single feedback loop, no need current information
- Good noise immunity with fixed ramp amplitude
- Fixed switching frequency
- Easy to work with a wide range of duty cycles

Disadvantage

- Type III compensation is necessary to support a wide range of LC filter combinations
- Slow load transient response due to the delay of EA and type III compensation
- Slow line transient response due to no use V_{in} information
- Additional solution to improve load/line transient response
 - I. Feed-forward
 - II. API(Asynchronous pulse injection)
 - III. AVP(adaptive voltage position/droop)
 - IV. ...

Peak Current Mode Control (PCM)



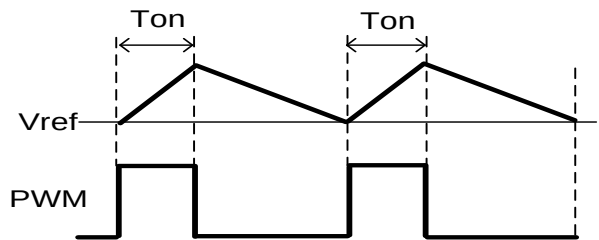
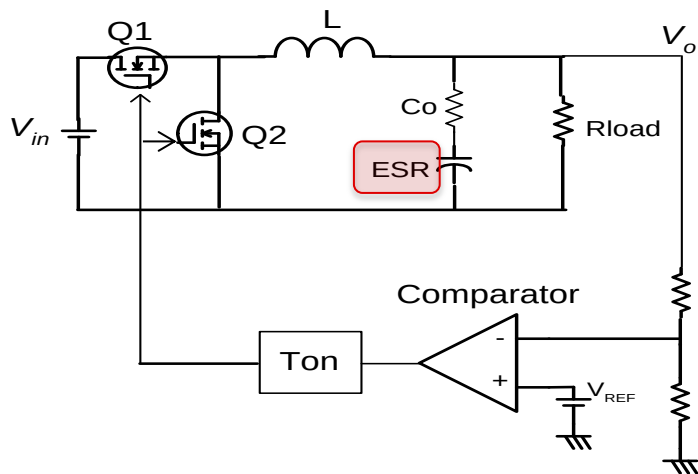
Advantage

- Fast load transient performance
- Single-pole system allows simple Type 2 compensation
- Inherent feed forward improves line transient performance
- Fixed switching frequency
- Easy implementation of cycle-by-cycle current limit
- Easy current share across multiple converters

Disadvantage

- Noise sensitivity at leading edge spike
- Need slope compensation to eliminate sub-harmonic oscillation
- Relatively long minimum on-time due to leading edge blanking

COT(constant on-time) Control



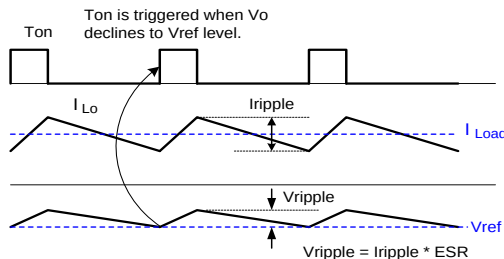
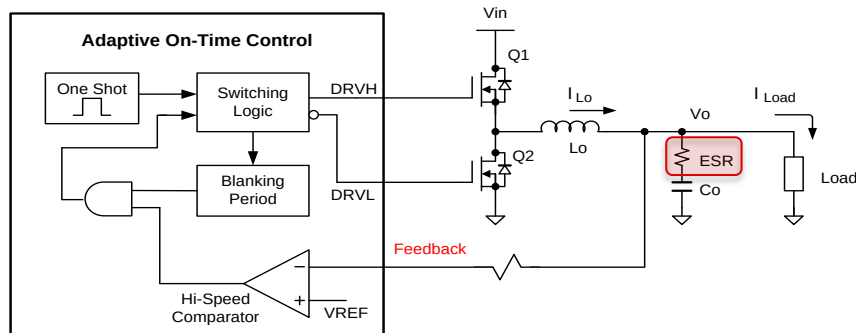
Advantage

- No loop compensation
- Fast load/line transient
- Inherent high light load efficiency

Disadvantage

- Variable switching frequency
- Requires some ESR output ripple
- Half-ripple offset on the output voltage
- Sensitive to output noise
- Relatively long minimum off-time due to valley output ripple sensed

D-CAP™ Control Mode (Adaptive On-time)



Direct Connect to the Capacitor (D-CAP)

- Each T_{on} pulse duration is calculated based upon V_{in} , V_{out} , nominal f_{out}
- Each time the falling feed-back voltage equals V_{ref} , a new ON pulse is generated
- There is no loop lag time. Pulse by pulse adjustment, comparator and the 'one shot block' lag are smaller

Advantage

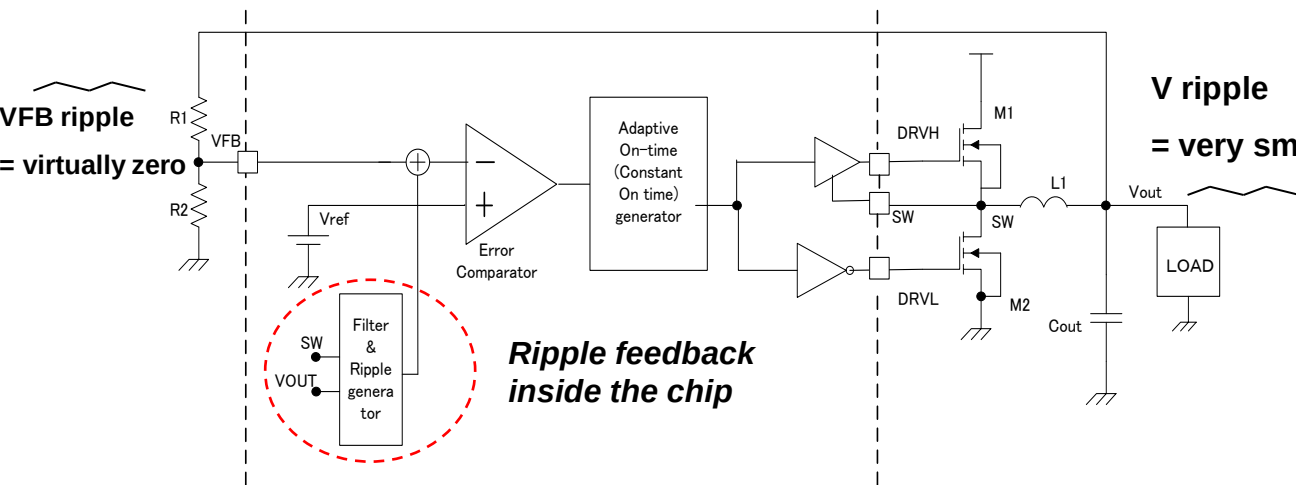
- No loop compensation
- Quasi fixed switching frequency
- Fast load/line transient performance
- Inherent high light load efficiency

Disadvantage

- T_{on} and T jitter is expected, with smaller L and C
- Requires some ESR output ripple
- Load Regulation
- Sensitive to output noise
- Relatively long minimum off-time due to valley output ripple sensed

D-CAP2™ Control Mode

Basically, D-CAP2™ is D-CAP™ plus **internal ripple injection** allowing for output ceramic capacitors use without added components or **ESR**



Advantage over DCAP

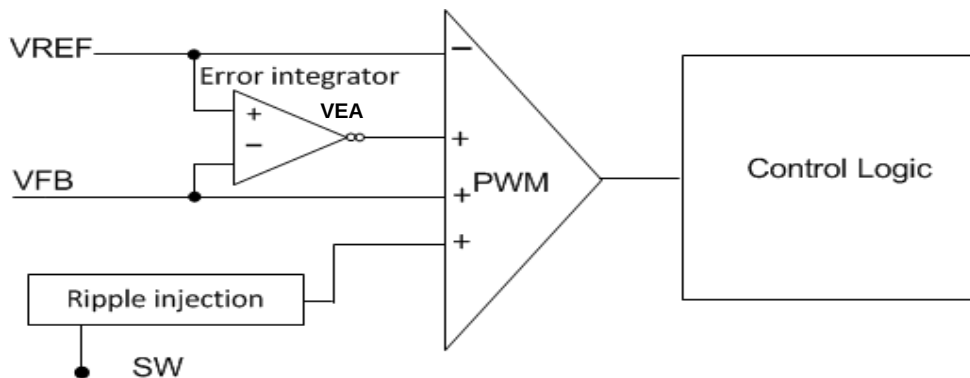
V ripple

= very small

- Use when ceramic output caps are required: Less jitter but output voltage accuracy is degraded by emulated current information.

D-CAP3™ Control Mode

Basically, D-CAP3™ is D-CAP2™ plus **internal integrator** allowing for eliminating the DC offset of output voltage

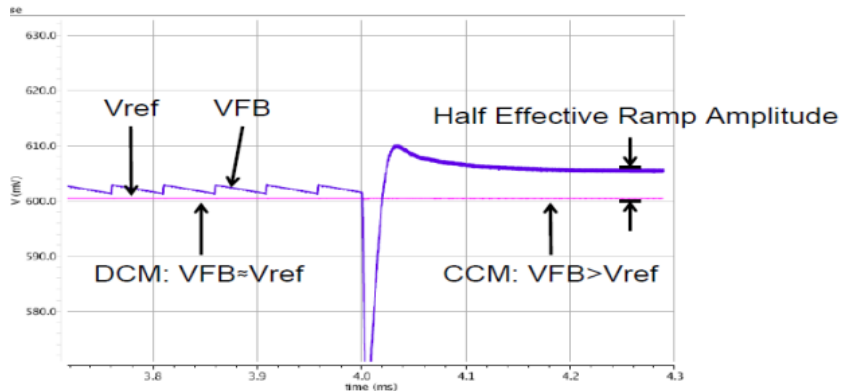


Advantage over DCAP2

- Improves output voltage accuracy

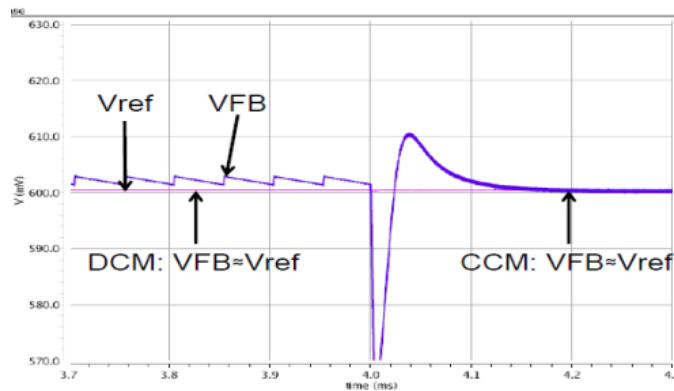
Load Transient Comparison Between DCAP2 & DCAP3

DCAP2



- In DCM, $V_{FB} \approx V_{ref}$
- In CCM, $V_{FB} = V_{ref} + \text{Ramp}/2 > V_{ref}$
- Offset generated by ramp reduces V_{out} accuracy

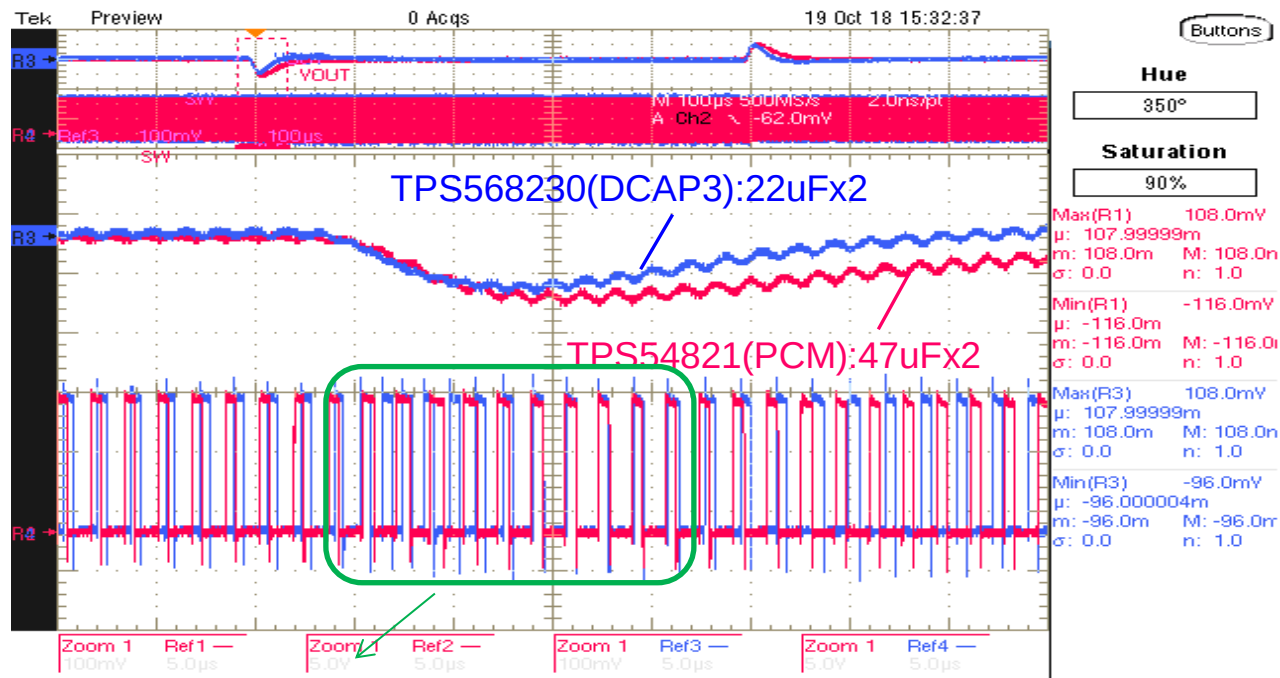
DCAP3



- In both DCM and CCM, $V_{FB} \approx V_{ref}$
- Offset generated by Ramp is removed and V_{out} accuracy is improved

Load Transient Comparison Between PCM & DCAP3

VIN=12V VOUT=3.3V IOU=0A-6A 2.5A/us Zoomed in



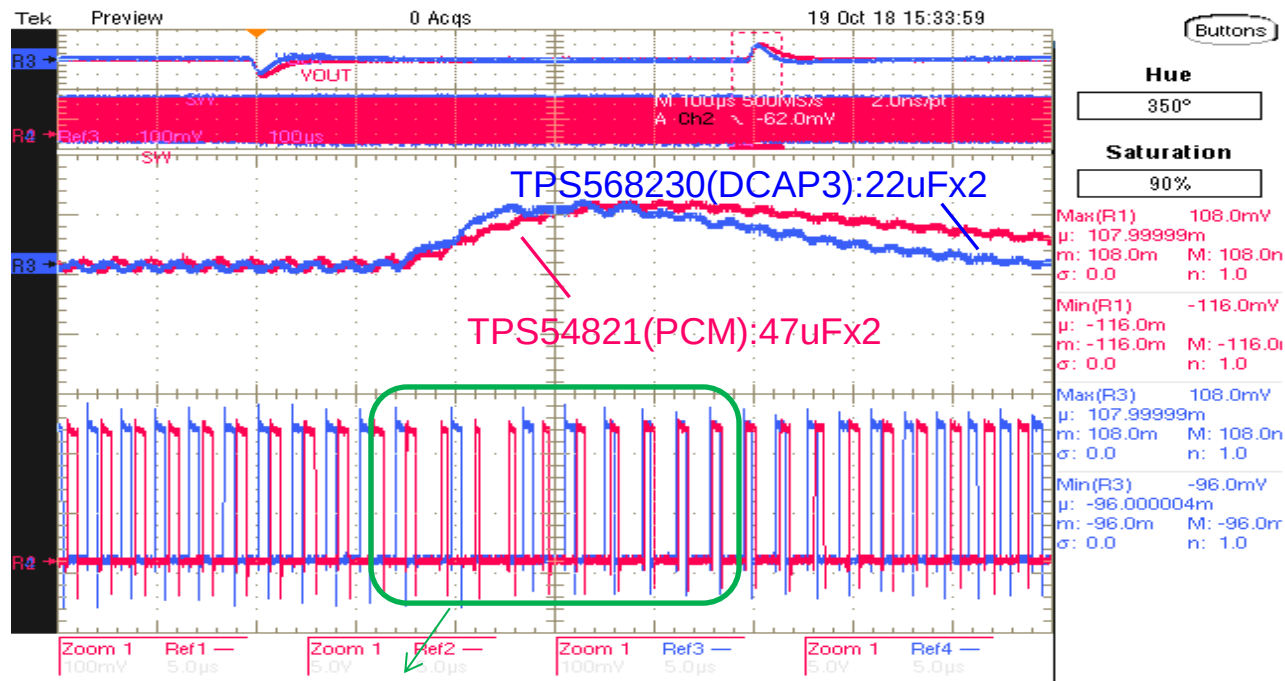
In DCAP3, during transient, Fsw increases to reduce undershoot.

In PCM, Fsw keeps constant during transient.

DCAP3 can save you 50uF!

Load Transient Comparison Between PCM & DCAP3

VIN=12V VOUT=3.3V IOU=6A-0A 2.5A/ μ s Zoomed in



In DCAP3, during transient, Fsw decreases to reduce overshoot.

In PCM, Fsw keeps constant during transient.

DCAP3 can save you 50uF!

High Level “Best to Use When” Summary

Voltage Mode

- Synchronized / fixed switching frequency is needed.
- Usable for small input voltage variations and lighter loads.

Peak Current Mode

- Synchronized / fixed switching frequency is needed with lower parts count.
- Easier to design with type II compensator – no double complex pole..

High Level “Best to Use When” Summary(cont.)

DCAP

- Fast transient response when using POSCAP or medium ESR Cout
- Quasi fixed Fsw.
- No loop-compensation calculation or components are needed

DCAP2

- Fast transient response with low-ESR ceramic output capacitors
- Quasi fixed Fsw
- No loop-compensation calculation or components are needed

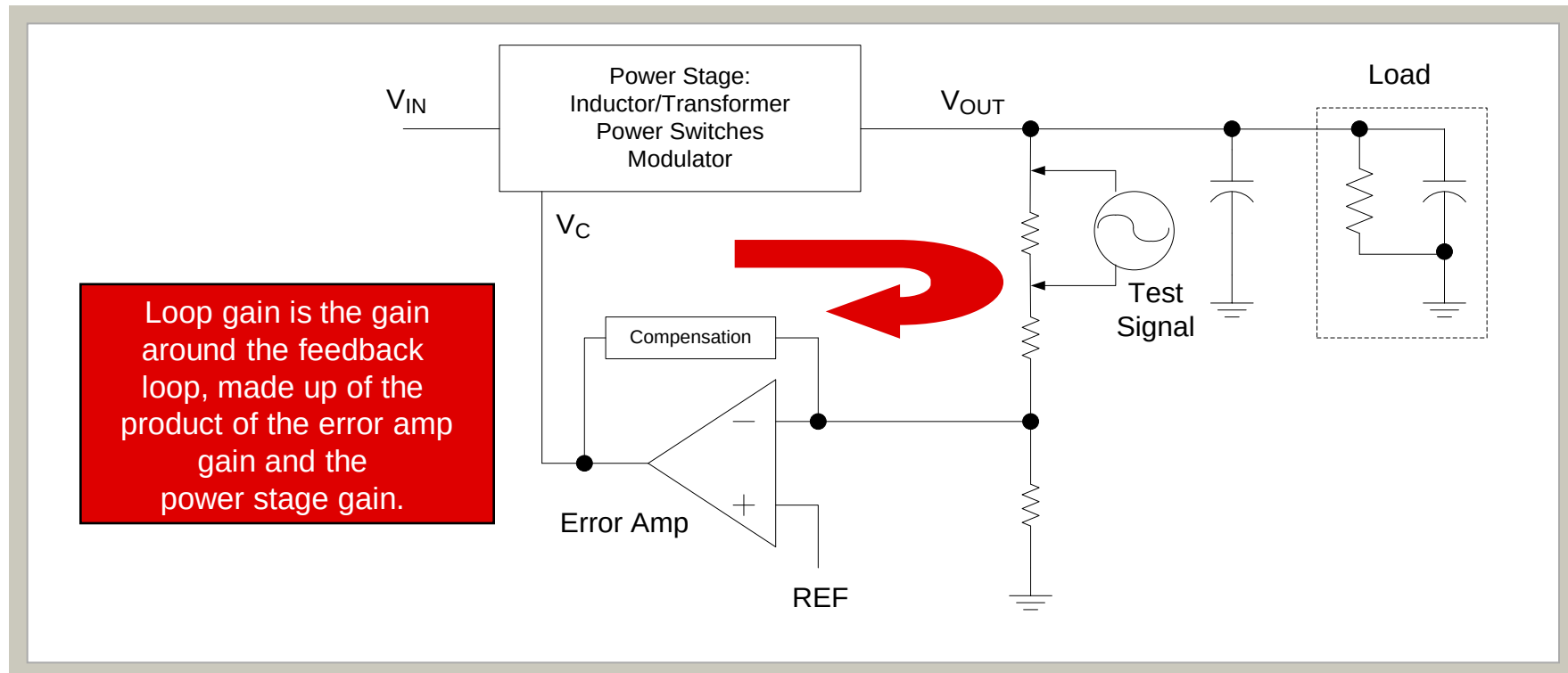
DCAP3

- Tighter reference voltage accuracy
- Fast transient response with low-ESR ceramic output capacitors
- Quasi fixed Fsw
- No loop-compensation calculation or components are needed

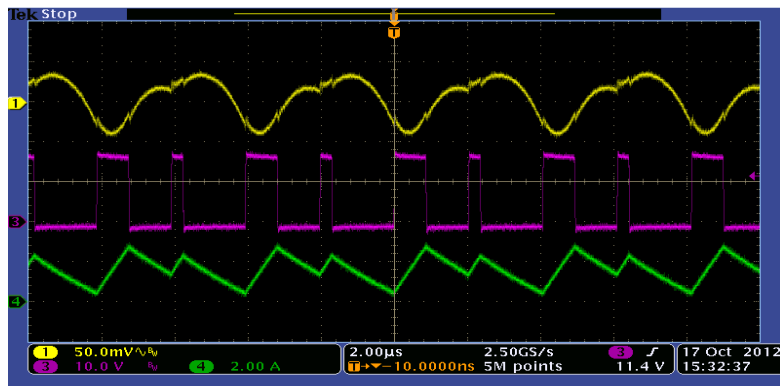
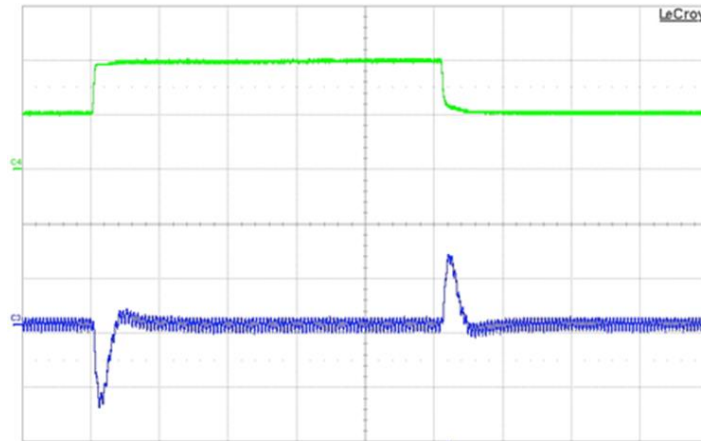
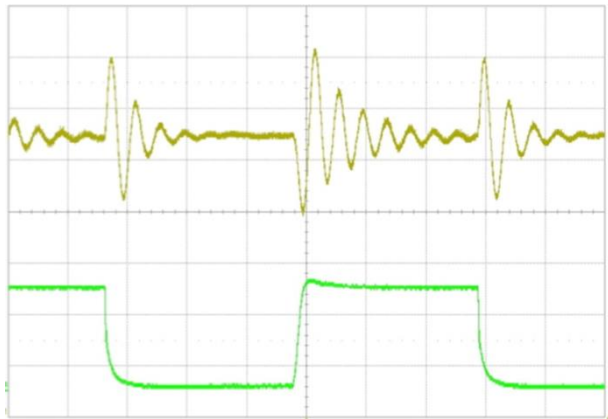
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Loop Compensation Introduction



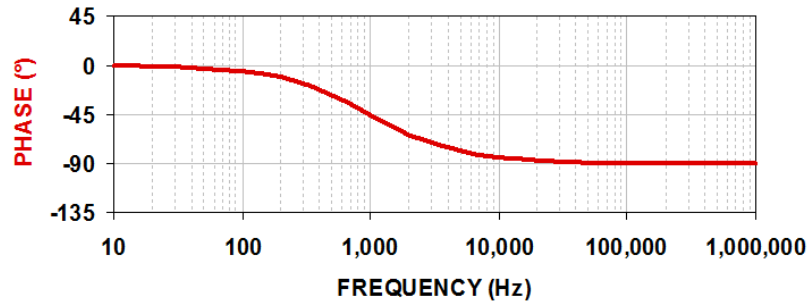
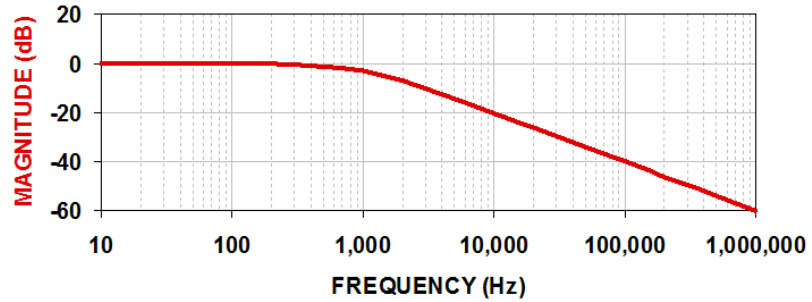
Why Do We Compensate?



- Bad compensation will cause worse transient performance.
- The loop may be unstable in the worst case, SW has a big jitter and output voltage has a big ripple.

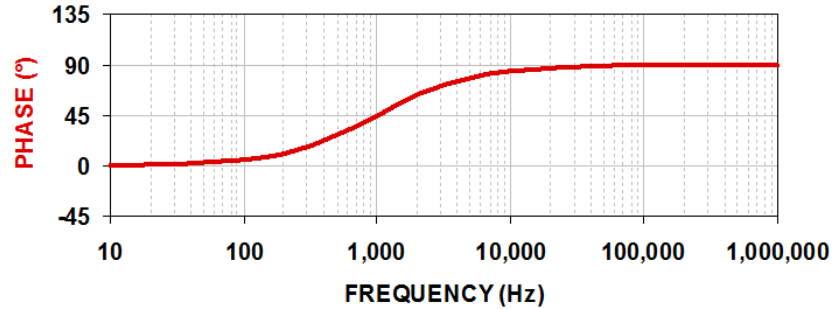
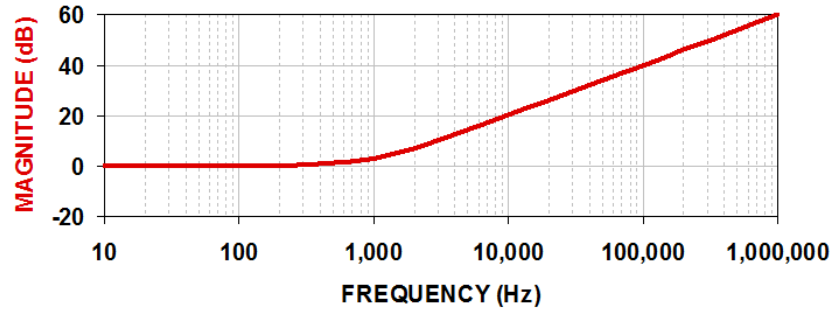
Single Pole

$$H(s) = \frac{1}{1 + \frac{s}{\omega_p}}$$

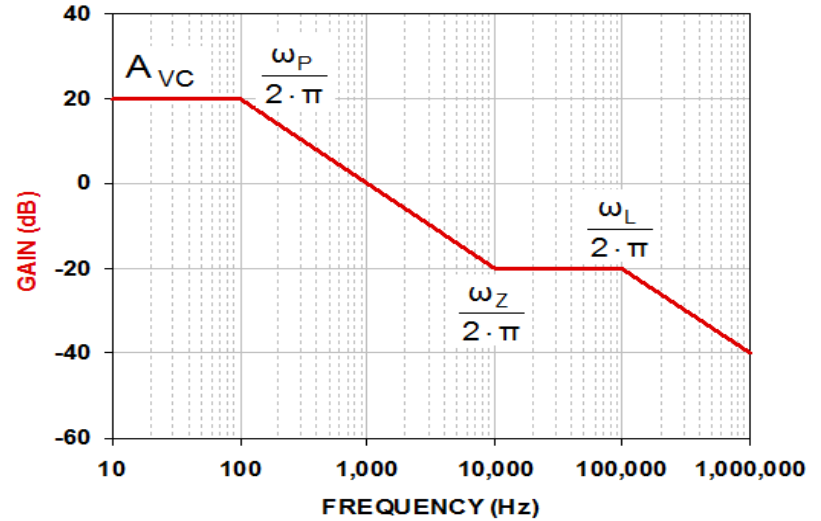
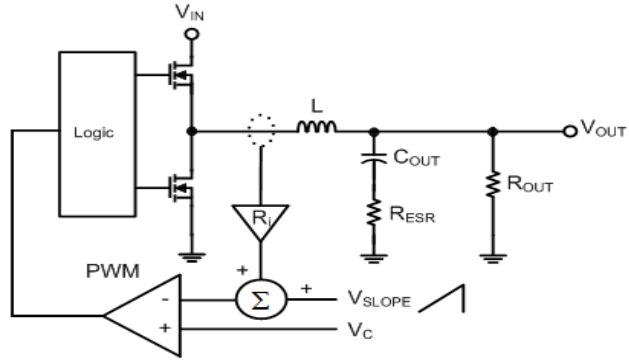


Single Zero

$$H(s) = \frac{1 + \frac{s}{\omega_z}}{1}$$



Current-Mode Buck Power Stage



$$A_{VC} \approx \frac{R_{OUT}}{R_f}$$

$$\omega_P \approx \frac{1}{C_{OUT} \cdot R_{OUT}}$$

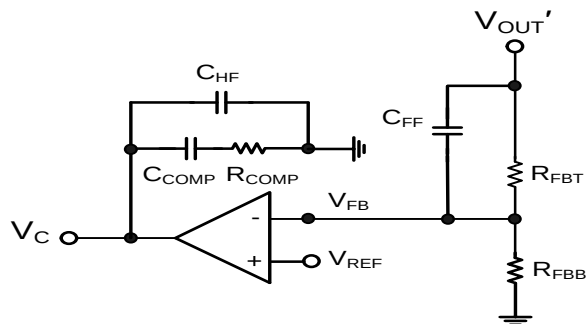
$$\omega_Z = \frac{1}{R_{ESR} \cdot C_{OUT}}$$

$$K_m \approx \frac{V_{IN}}{V_{SLOPE}}$$

$$\omega_L = \frac{K_m \cdot R_f}{L}$$

$$\frac{\hat{v}_{OUT}}{\hat{v}_C} \approx A_{VC} \cdot \frac{1 + \frac{s}{\omega_Z}}{\left(1 + \frac{s}{\omega_P}\right) \cdot \left(1 + \frac{s}{\omega_L}\right)}$$

Type II Compensation with Feedforward Cap



$$A_{VM} \approx \frac{R_{COMP}}{R_{FBT}}$$

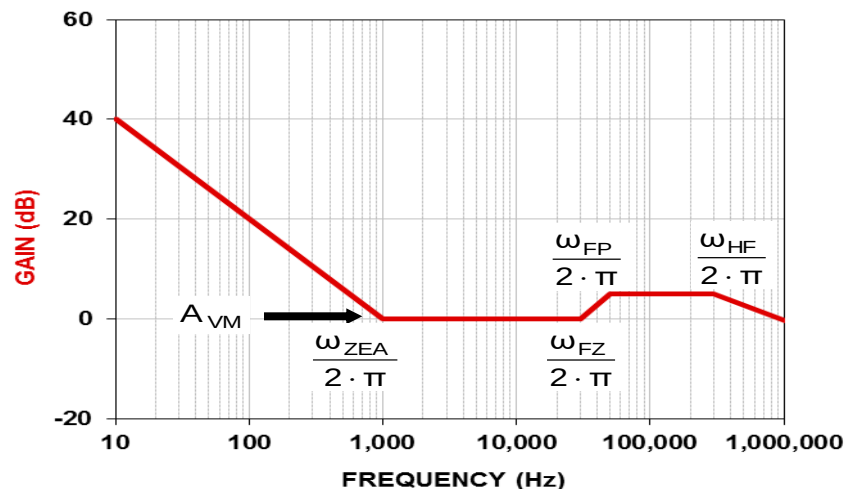
$$\omega_{ZEA} = \frac{1}{R_{COMP} \cdot C_{COMP}}$$

$$\omega_{FZ} \approx \frac{1}{R_{FBT} \cdot C_{FF}}$$

$$\omega_{FP} = \frac{1}{(R_{FBT} \parallel R_{FBB}) \cdot C_{FF}}$$

$$\omega_{HF} \approx \frac{1}{R_{COMP} \cdot C_{HF}}$$

Assumptions : $C_{COMP} \gg C_{HF}$ & $R_{FBT} \gg R_{FF}$

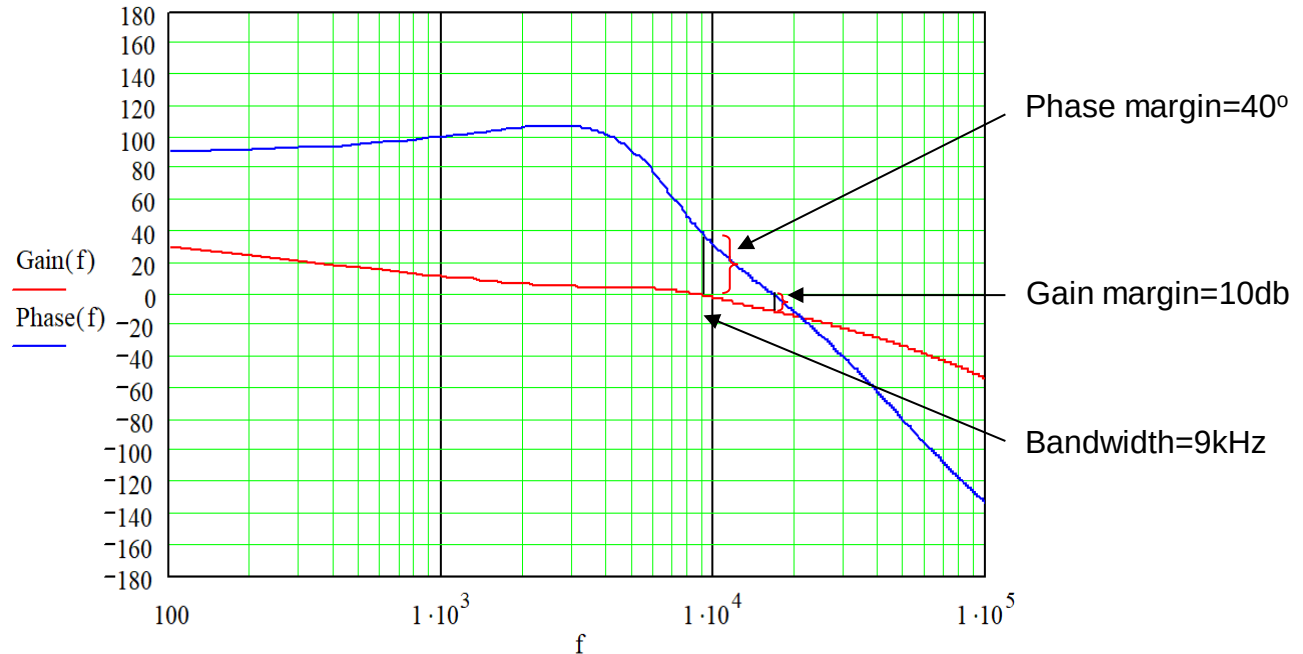


$$\frac{\hat{v}_{C'}}{\hat{v}_{OUT}} = -A_{VM} \cdot \frac{\left(1 + \frac{\omega_{ZEA}}{s}\right) \cdot \left(1 + \frac{s}{\omega_{FZ}}\right)}{s \cdot \left(1 + \frac{s}{\omega_{FP}}\right) \cdot \left(1 + \frac{s}{\omega_{HF}}\right)}$$

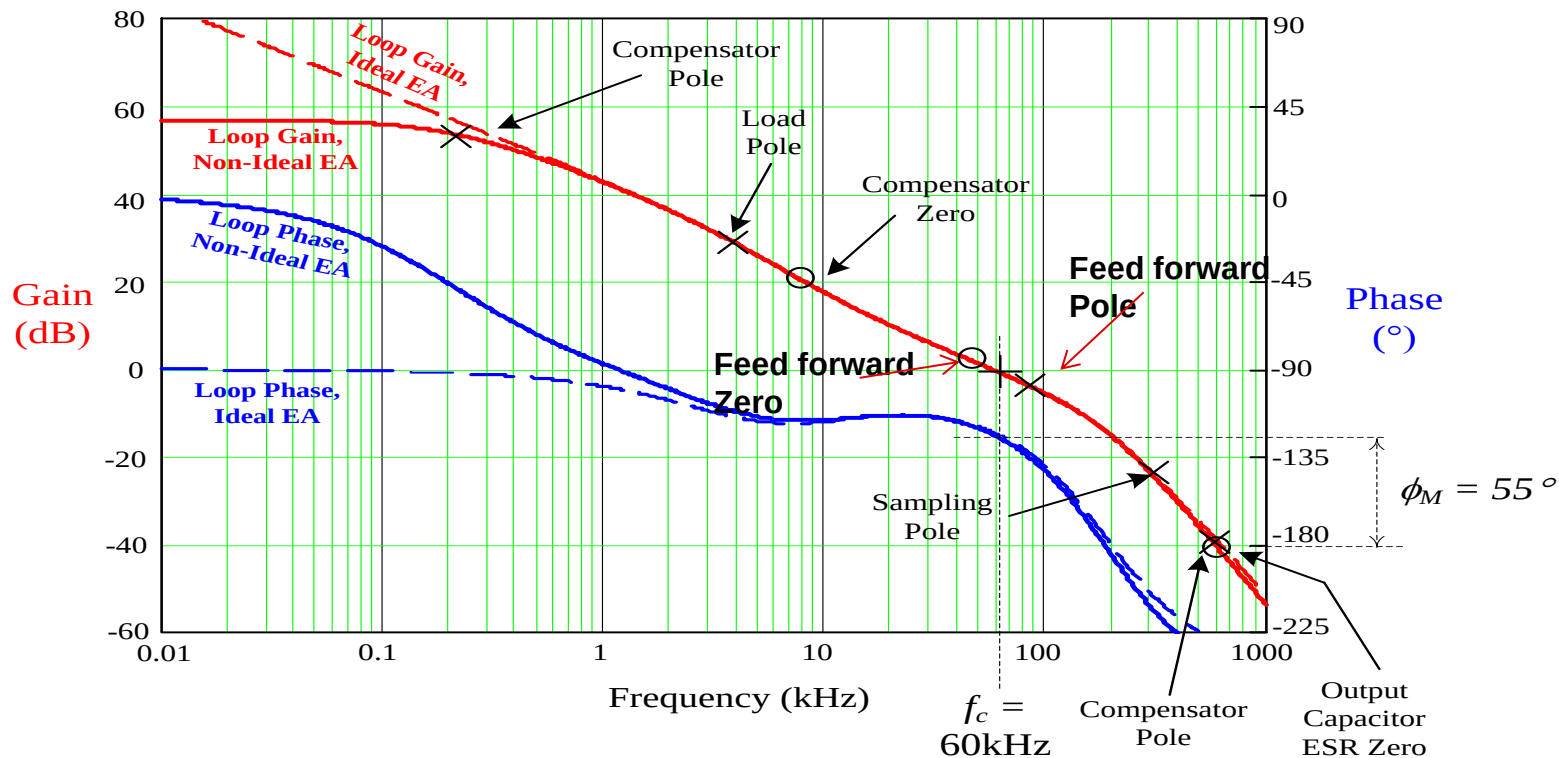
Loop Gain Bode Plot

The condition for good stability and fast dynamic response:

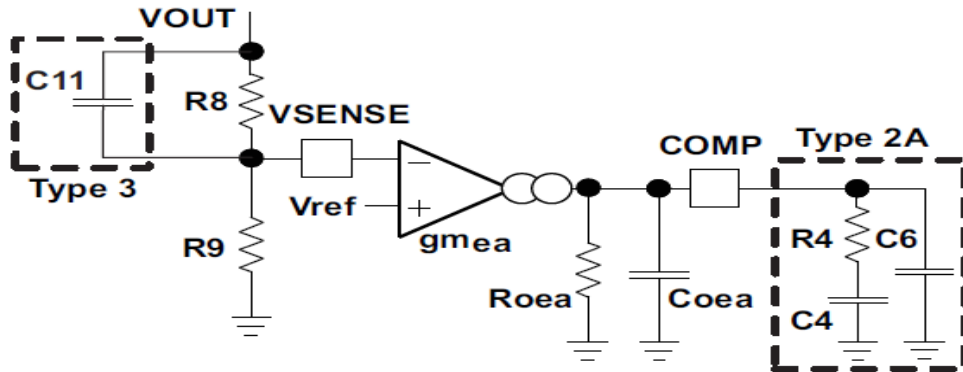
1. Phase margin should be larger than 45°
2. Gain margin should be larger than 6db;
3. Bandwidth is about 1/10 to 1/5 of switching frequency.



Loop Gain Bode Plot



How to calculate Compensation?



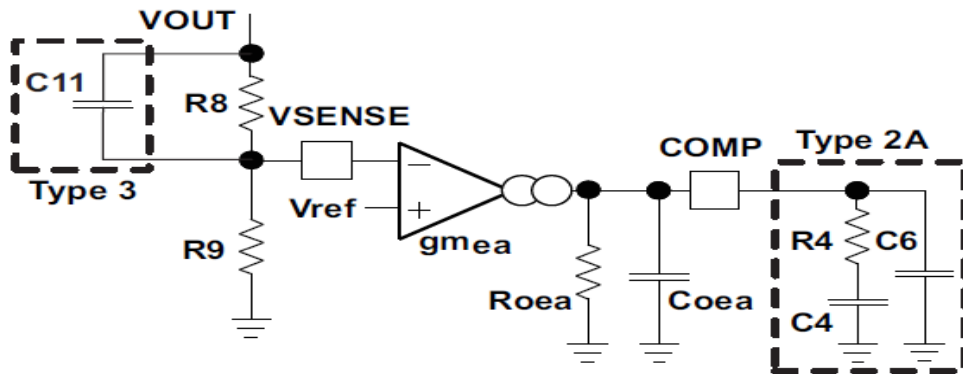
The general design guidelines for device loop compensation are as follows:

1. Determine the crossover frequency, f_c . A good starting point is 1/10th of the switching frequency, f_{sw} .
2. R_4 can be determined by:

$$R_4 = \frac{2\pi \times f_c \times V_{OUT} \times C_o}{g_{m_{ea}} \times V_{ref} \times g_{m_{ps}}}$$

Where: G_{mea} is the GM amplifier gain, G_{mps} is the power stage gain, V_{ref} is the reference voltage.

How to calculate Compensation?



3. Place a compensation zero at the dominant pole: $f_p = \frac{1}{C_o \times R_L \times 2\pi}$

C4 can be determined by:

$$C4 = \frac{R_L \times C_o}{R4}$$

4. C6 is optional. It can be used to cancel the zero from the ESR (Equivalent Series Resistance) of the output capacitor C_o.

$$C6 = \frac{R_{ESR} \times C_o}{R4}$$

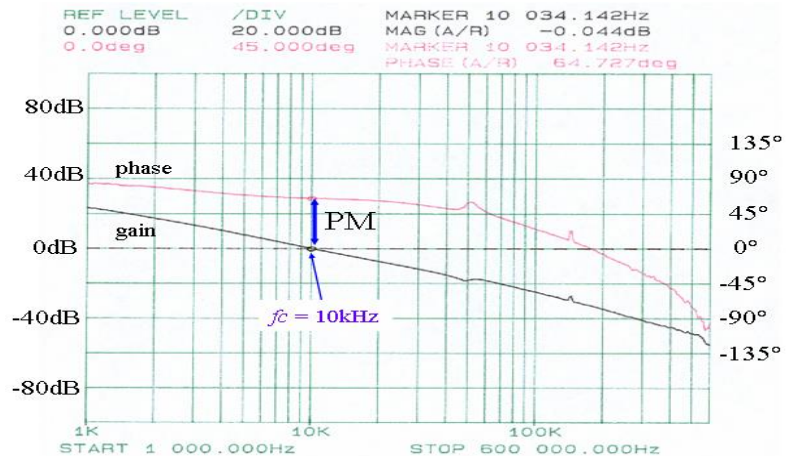
5. Type III compensation can be implemented with the addition of one capacitor, C11. This allows for slightly higher loop bandwidths and higher phase margins.

$$C11 = \frac{1}{(2 \cdot \pi \cdot R8 \cdot f_c)}$$

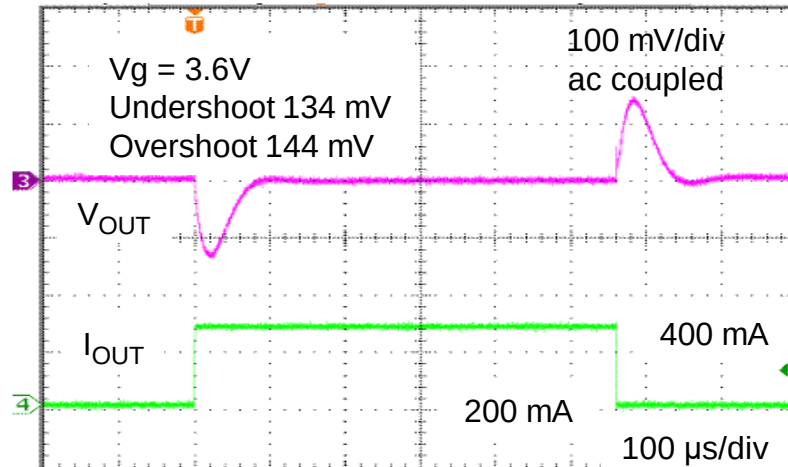
Bode Plot vs Transients

Case 1 – Stable Regulator

$$f_c = 10 \text{ kHz}, \text{ PM} = 65^\circ$$



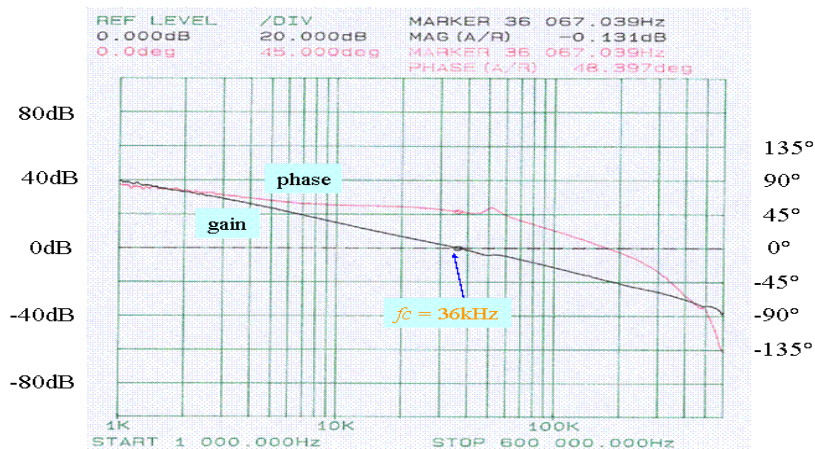
Over-Damped



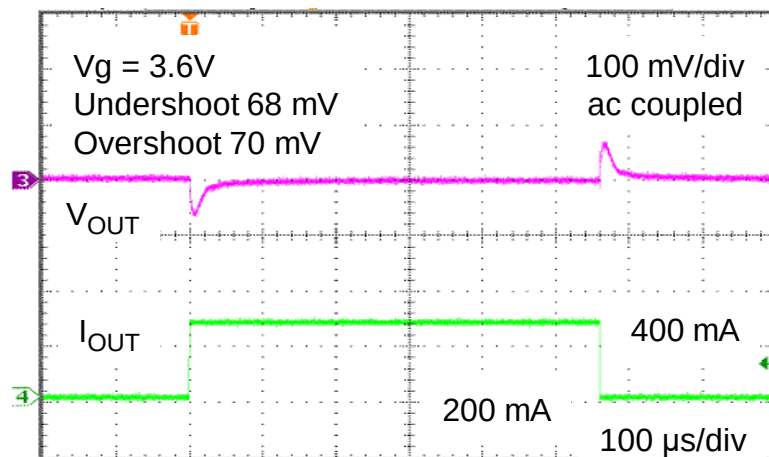
Bode Plot vs Transients

Case 2 – Stable Regulator

$f_c = 36 \text{ kHz}$, $PM = 48^\circ$



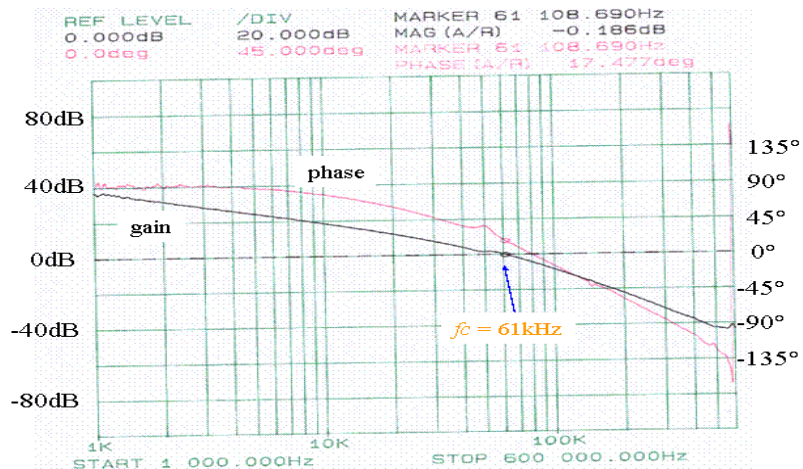
Critically-Damped



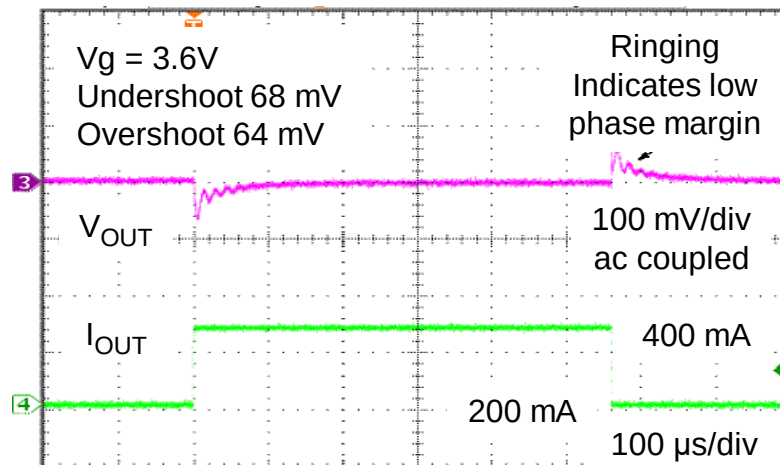
Bode Plot vs Transients

Case 3 – Marginal Stability

$$f_c = 61 \text{ kHz}, \text{ PM} = 17^\circ$$



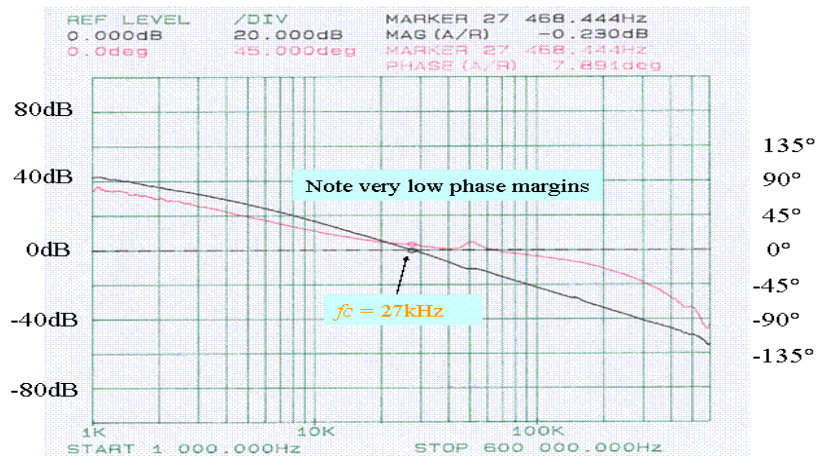
Under-Damped



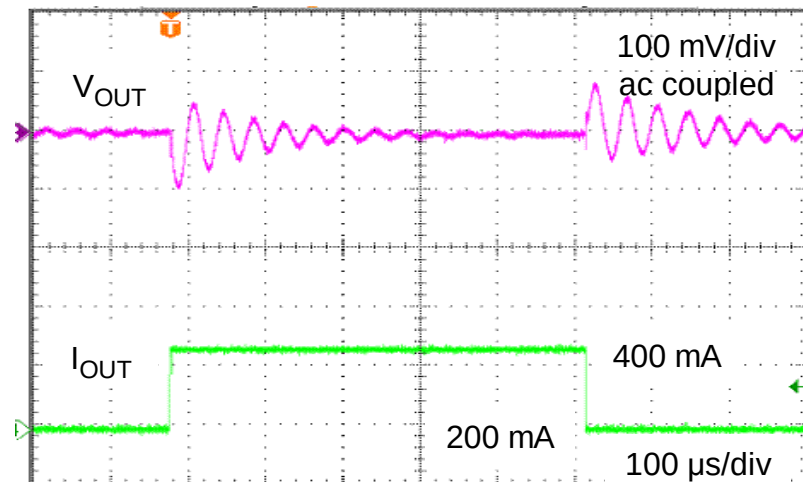
Bode Plot vs Transients

Case 4 – Unstable Regulator

$$f_c = 27 \text{ kHz}, \text{ PM} = 8^\circ$$



Unstable Regulator



Optimize Compensation

Bandwidth:

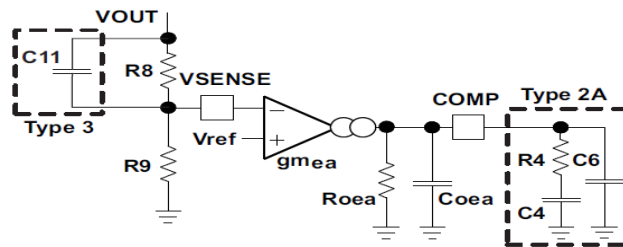
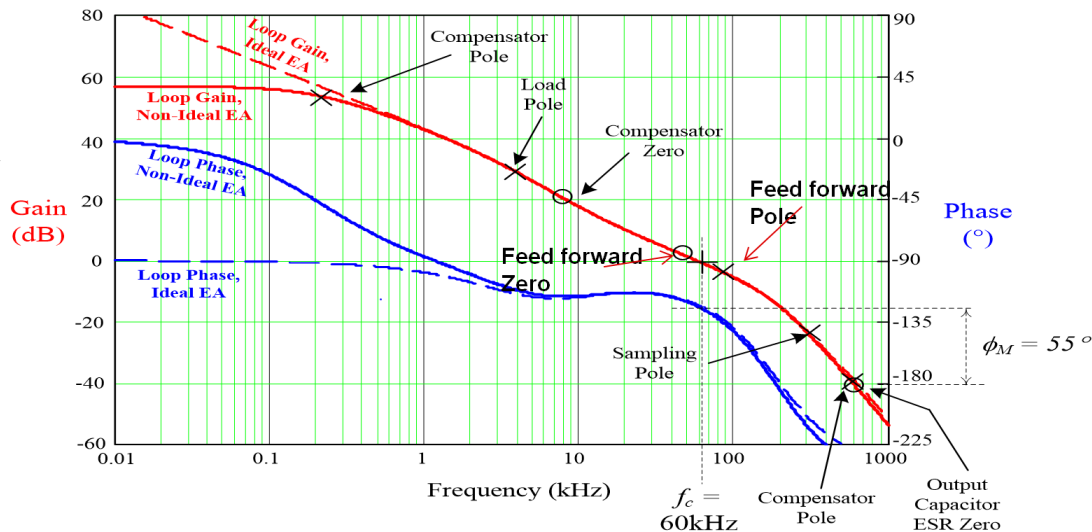
$$f_c = \frac{R_4 \times g_{m_{ea}} \times V_{ref} \times g_{m_{ps}}}{2\pi \times V_{out} \times C_{out}} \rightarrow R_4, C_{out}$$

Phase Margin:

$$f_z = \frac{1}{2\pi \times R_4 \times C_4} \rightarrow R_4, C_4$$

Gain Margin:

$$f_p = \frac{1}{2\pi \times R_4 \times C_6} \rightarrow R_4, C_6$$



Use Feed Forward Capacitor to Increase Phase Margin

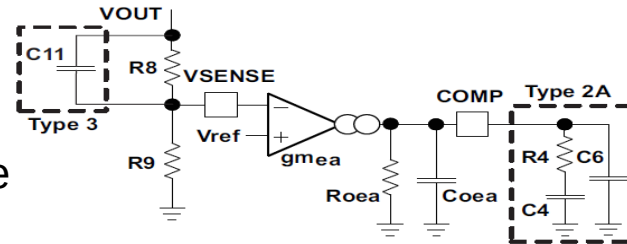
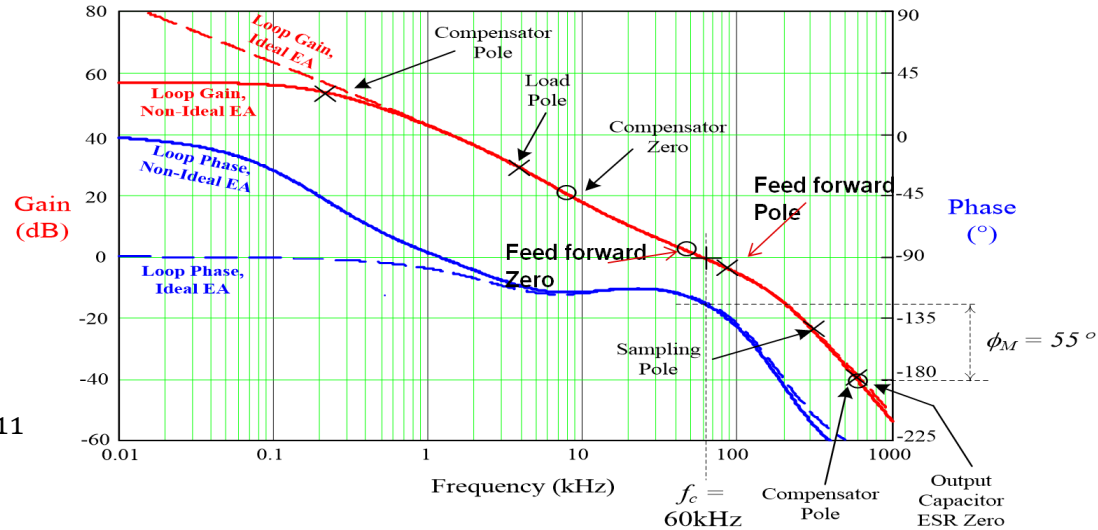
- Feed forward capacitor adds a zero / pole pair:

$$f_z = \frac{1}{2\pi \times R_8 \times C_{11}} \rightarrow R_8, C_{11}$$

$$f_p = \frac{1}{2\pi \times \frac{R_8 \times R_9}{R_8 + R_9} \times C_{11}} \rightarrow R_8, R_9, C_{11}$$

$$f_z < f_p$$

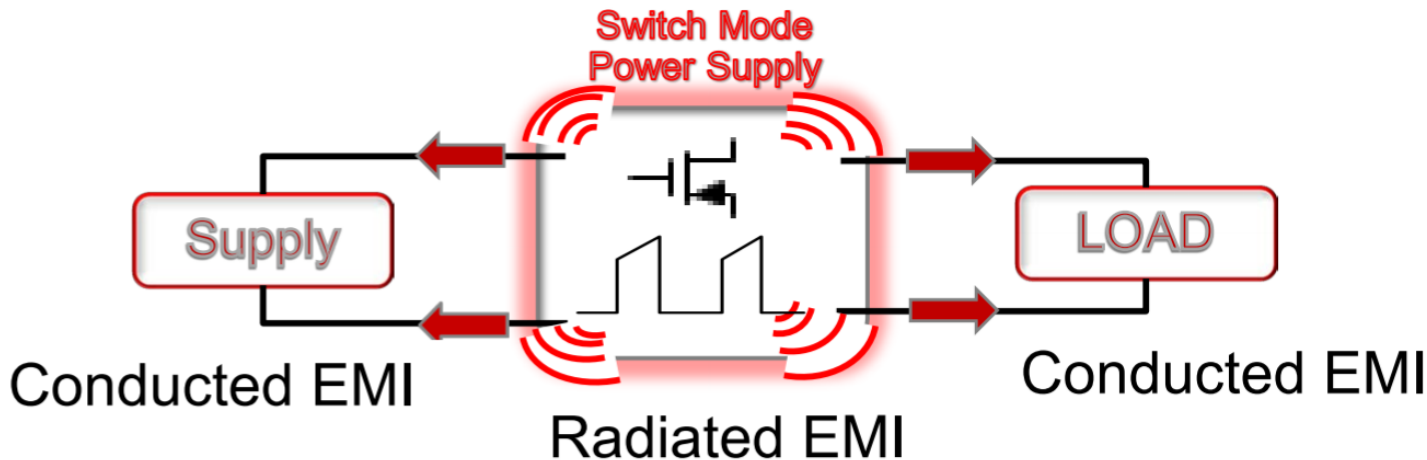
- Place the zero / pole pair close to f_c to increase phase margin.



Agenda:

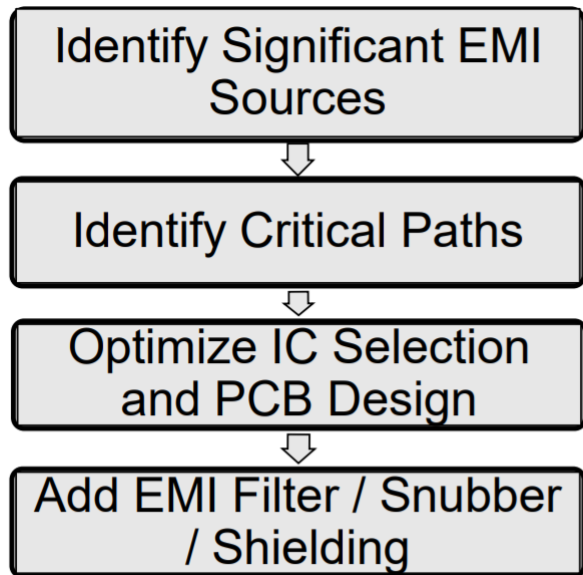
- 一、Buck 拓扑及功率器件选型介绍
- 二、Buck 变换器常见控制模式介绍
- 三、环路分析及补偿设计(PCM)
- 四、EMI 与 PCB 布板设计**
- 五、Buck 常见问题及解决方法

EMI in Switch Mode Power Supplies



GOAL: **EMI Noise Generation** ↓ **EMC Noise Compatibility** ↑

Steps To Mitigate EMI In PCB Design



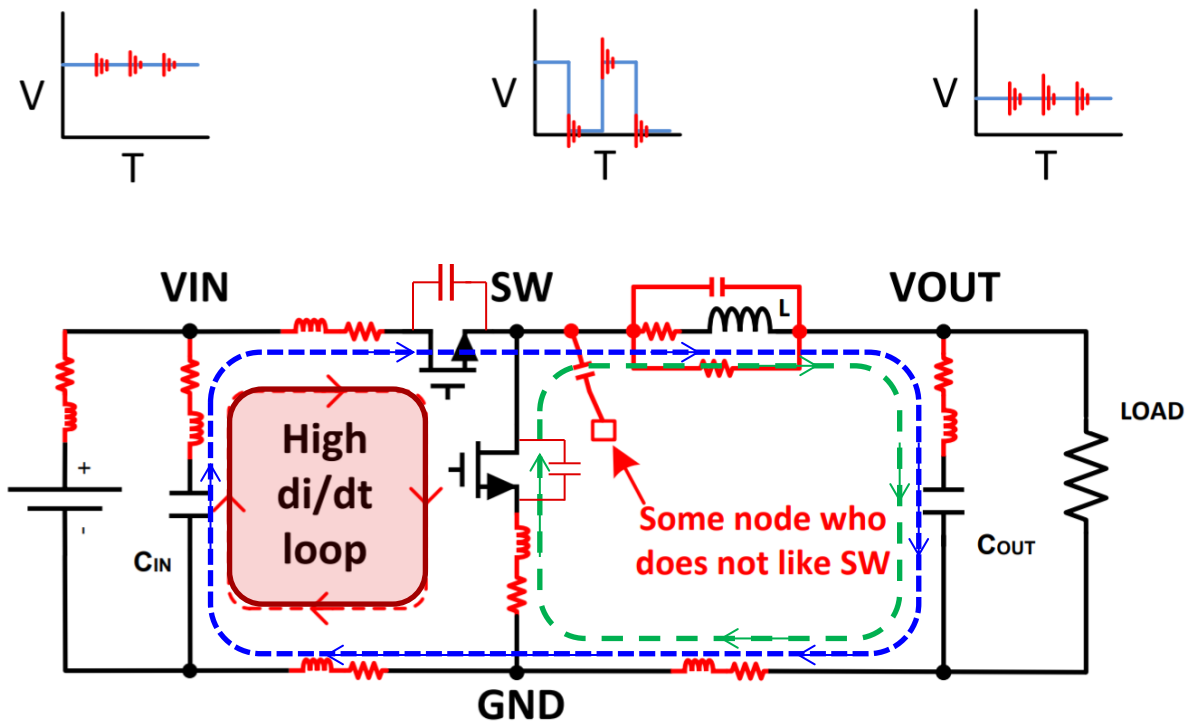
Switching components generate high di/dt current and high dv/dt voltage



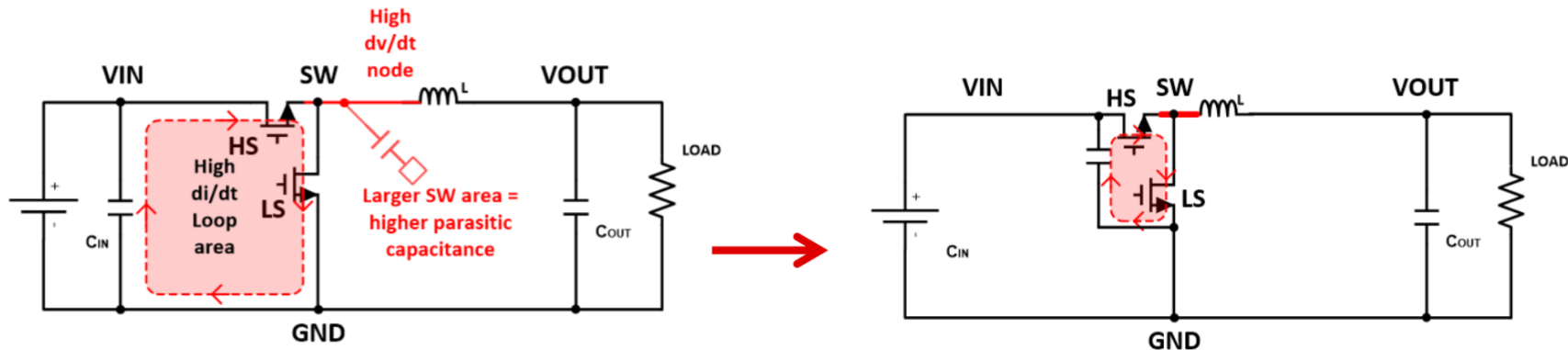
Loop Contains high di/dt current is **CRITICAL PATH;**

Node Contains high dv/dt voltage is **CRITICAL NODE.**

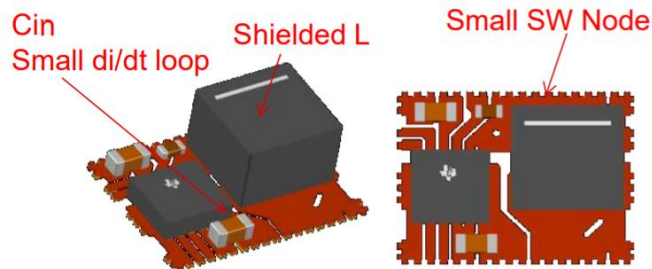
Less Ideal Buck Converter



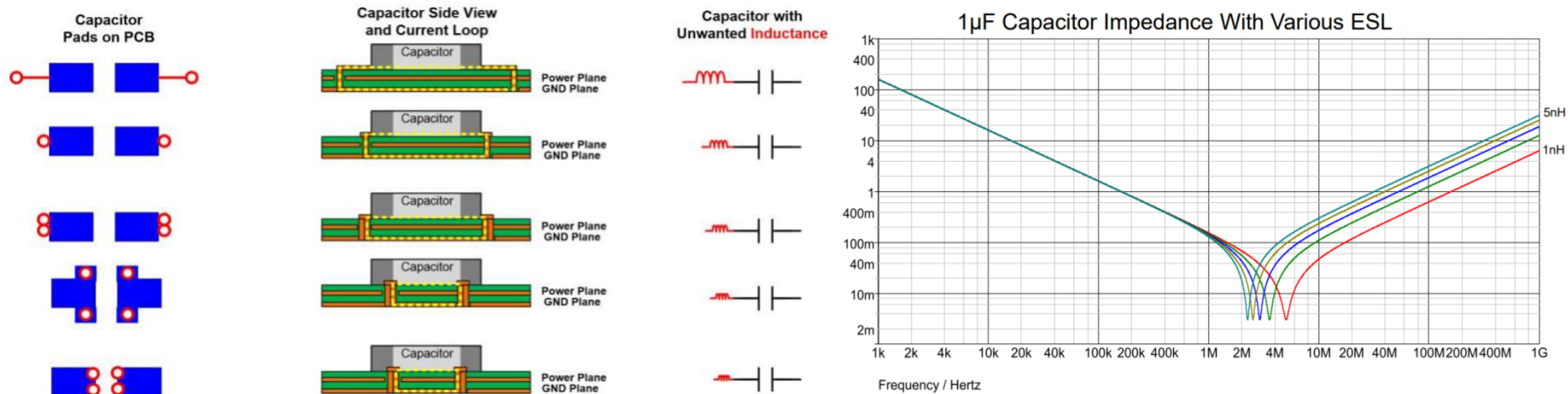
What can we do?



- ✓ Reducing the high di/dt loop area: Place input capacitor **as close to pin as possible**
- ✓ Reducing the high dv/dt node area: Keep SW trace **short and wide**

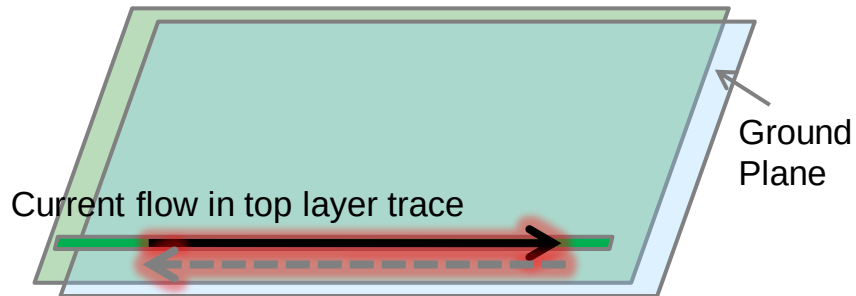


Bypass capacitor routing example

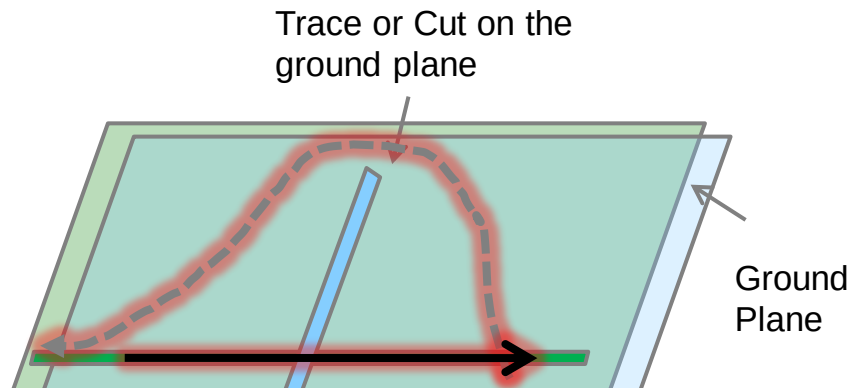


- ✓ Place bypass capacitors on same side of board as component being decoupled
- ✓ Locate as close to pin as possible
- ✓ Keep trace width thick and short

Ground Plane



Return current path in unbroken ground plane directly under path
Area minimized

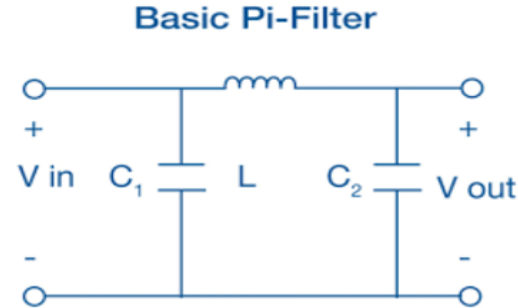
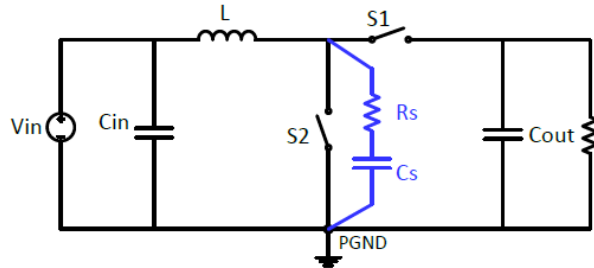


Return current path enclose much larger area if the direct path is blocked

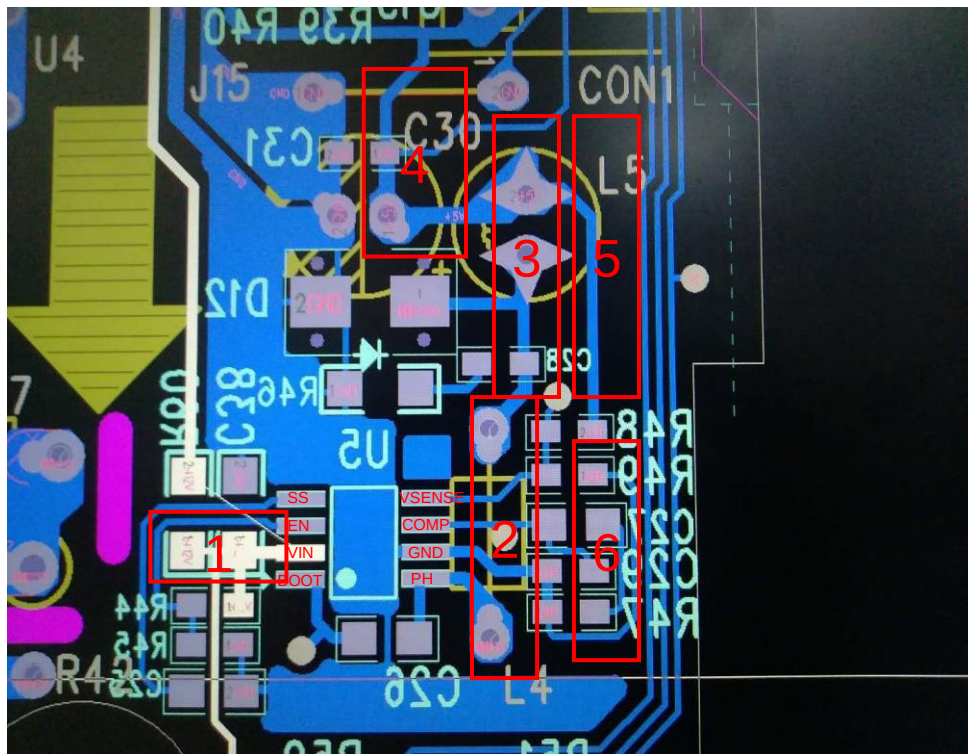
Better to use unbroken ground plane

Other method

- Snubber (radiant-EMI)
- Pi-filter as input filter
- Ferrite Bead
 - A ferrite bead or ferrite choke is a passive electric component that suppresses high frequency noise in electronic circuits. It is a specific type of electronic choke.



DC-DC PCB 布板原则



基本原则:

1. 开关高频电流环路尽量短
2. 开关高频电压结点尽量小
3. 大电流环路尽量短
4. 反馈电阻尽量靠近FB脚
5. 反馈点尽量靠近输出端口处
6. 信号地和大电流地单独走线，并在芯片GND脚单点连接

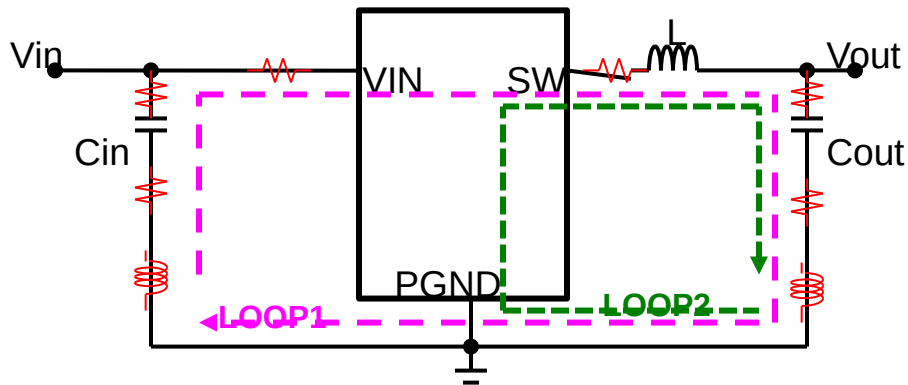
不好的布板会引起:

1. EMI
2. 地线噪声
3. 布线上产生电压降
4. 使环路调节能力下降，甚至使环路不稳

开关高频电流环路及大电流环路尽量短

• How

- 输入电容尽量靠近 VIN脚
- 电感尽量靠近SW脚
- 输入输出电容的接地端尽量靠近 PGND 脚
- 使用铺铜的方式走线



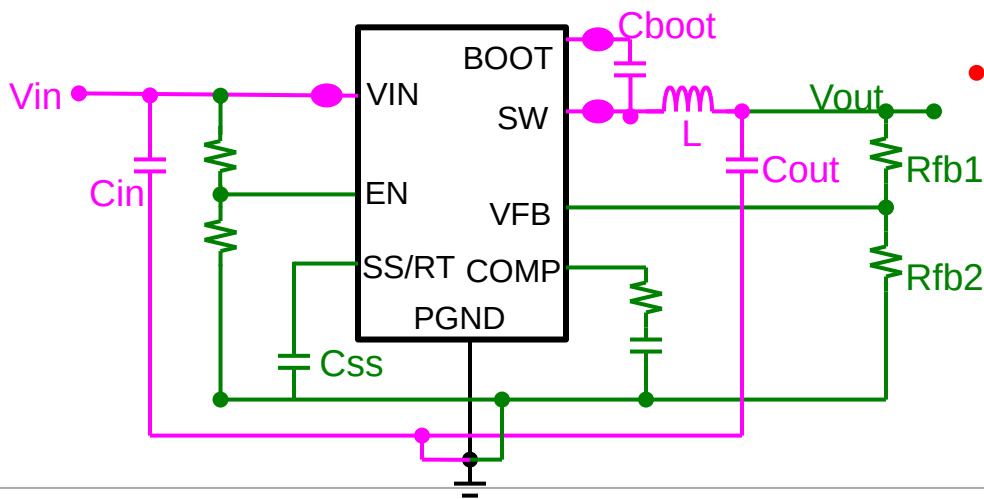
• Why

- 高频电流环路过大会导致EMI及纹波噪声问题
- 电感结点面积过大会导致EMI问题
- 走线过细过长会增大其电阻，这样开关大电流会在此电阻上产生纹波电压
- 走线过细过长会增大其寄生电感，此电感上会耦合开关噪音，产生电压尖刺，有可能会干扰芯片正常的驱动脉冲输出，影响环路稳定性，并造成电磁干扰EMI问题
- 寄生电容和电阻会增大开关及导通损耗，影响效率

大电流开关网络和小信号网络要分开

• How

- 大电流地: L, Cin, Cout, Cboot 连接到大电流地的网络
- 小信号地: C_{SS}, R_{RT}, 反馈电阻 Rfb1 和 Rfb2 单独连接到信号地的网络



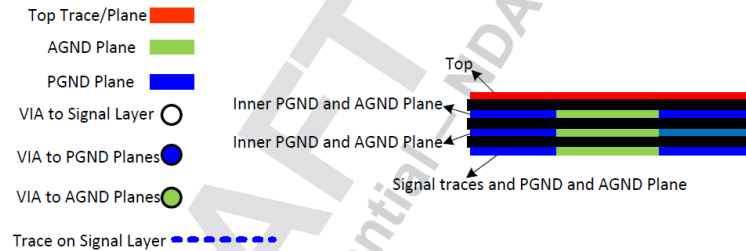
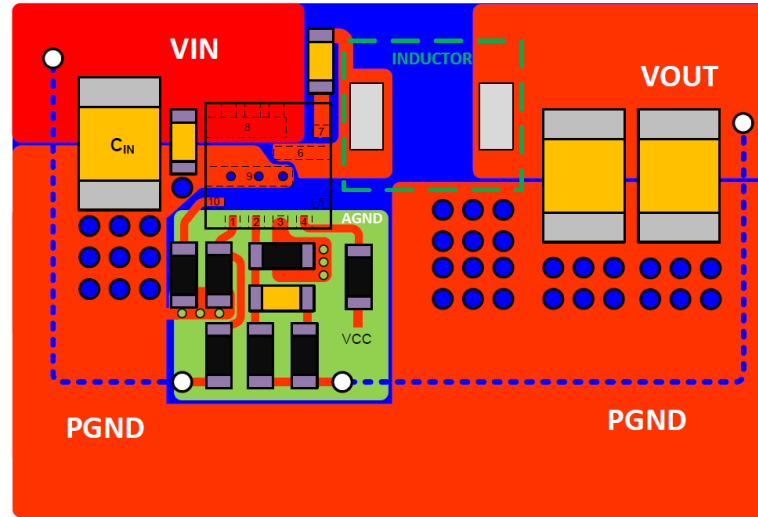
• Why

- 尽量减少大电流开关噪声对敏感的小信号网络的干扰.
- 例如: $V_{FB} = V_{out} / (R_{fb1} + R_{fb2}) * R_{fb2}$; V_{out} 是按 V_{FB}=V_{REF} 的条件设置的分压电阻网络; 如果 V_{FB} 上耦合了较多的噪音, 影响了其直流电平, 则 V_{out} 就会和预期的输出值有偏差.

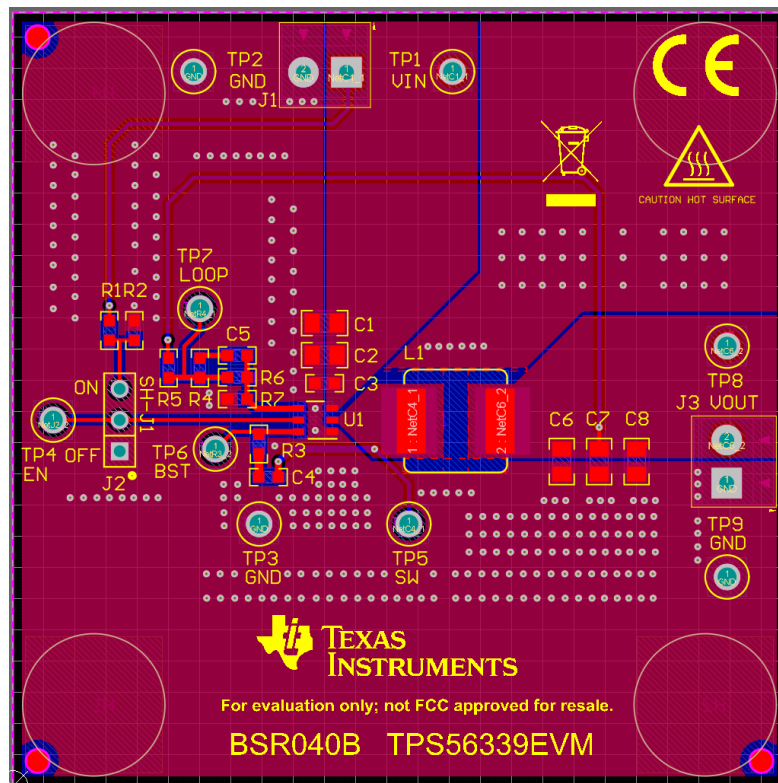
• If

- 如果因为空间限制, 这两个网络不能完全分开走线的话, 可以遵循如下原则
 - 两个网络的元件放置在不同面;
 - 使用地线屏蔽
 - 尽量采用垂直走线方式, 避免两个网络平行走线

Layout Example - QFN

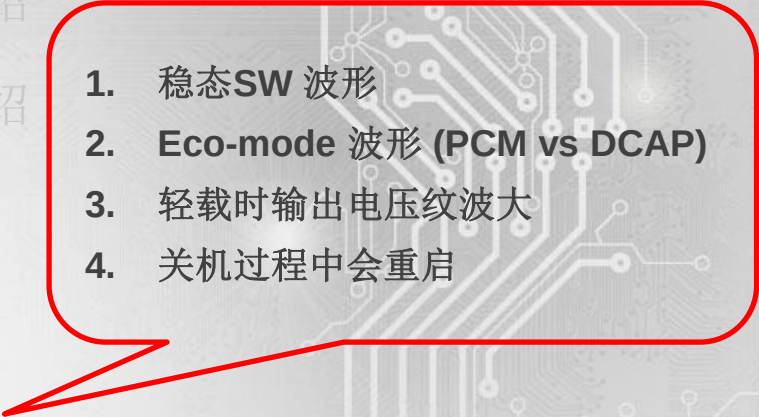


Layout Example – SOT23-6



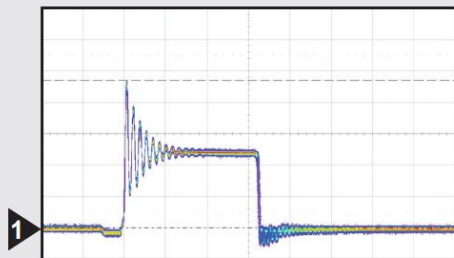
Agenda:

- 一、Buck 拓扑及功率器件选型介绍
- 二、Buck 变换器常见控制模式介绍
- 三、环路分析及补偿设计(PCM)
- 四、EMI 与 PCB 布板设计
- 五、**Buck常见问题及解决方法**

- 
- 1. 稳态SW 波形
 - 2. **Eco-mode 波形 (PCM vs DCAP)**
 - 3. 轻载时输出电压纹波大
 - 4. 关机过程中会重启

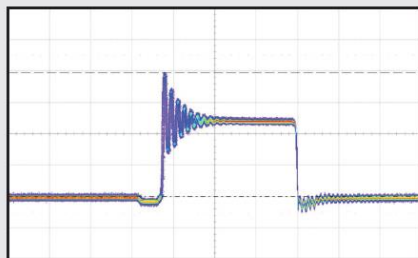


稳态SW 波形



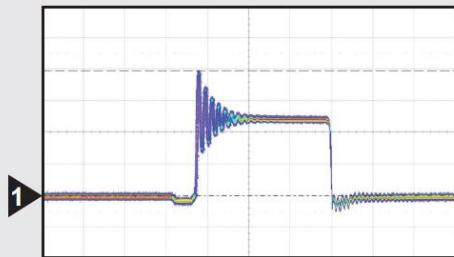
Time (50 ns/div)

(a) Baseline ringing (5 V/div)



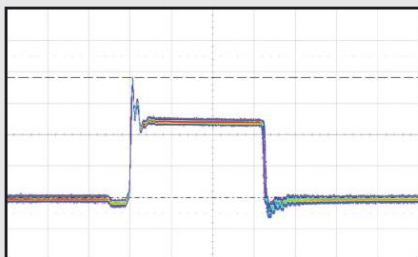
Time (50 ns/div)

(b) Ringing with boot resistor (5 V/div)



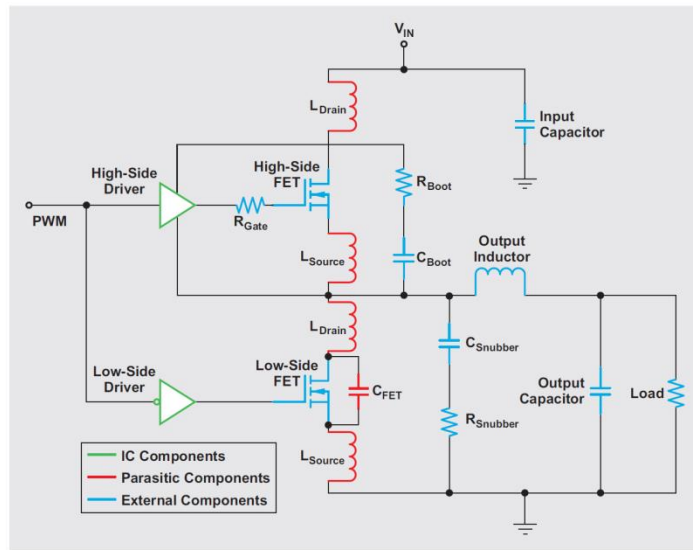
Time (50 ns/div)

(c) Ringing with high-side gate resistor (5 V/div)



Time (50 ns/div)

(d) Ringing with snubber (5 V/div)

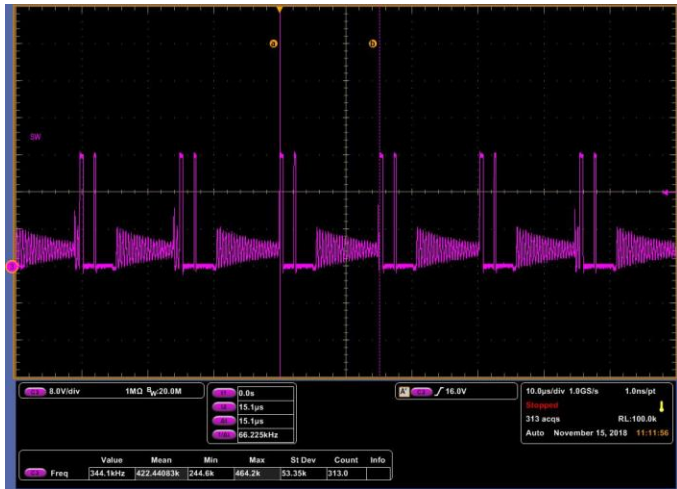


减小SW振铃的方法:

1. 增大Rboot 或 Rgate
 2. 加吸收电路
 3. 优化layout (将输入电容尽量靠近芯片VIN和GND引脚, 并且在输入端加0.1uF陶瓷电容)
- 缺点: 效率降低

Eco-mode 波形 (PCM vs DCAP)

PCM



DCAP



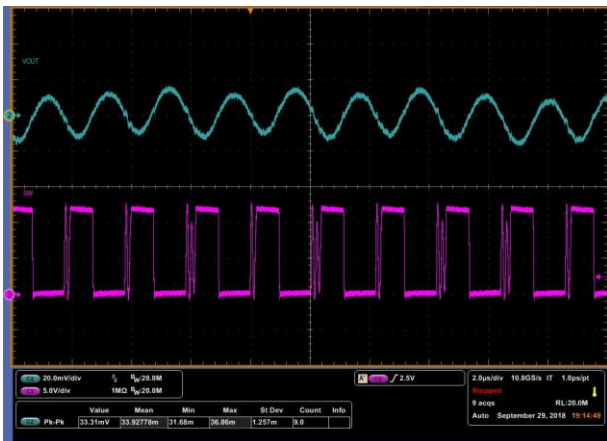
- 对于DCAP控制的器件，它是恒定导通时间控制，可以通过调节关断时间来改变开关频率。所以它可以实现单脉冲。
- 但是对于PCM控制的器件，时钟脉冲是恒定的，只能通过跳脉冲来改变开关频率。而进/出跳频模式是由Vcomp决定的，环路的响应也需要一定的时间，所以很难实现有规律的单脉冲控制。

轻载时输出电压纹波大

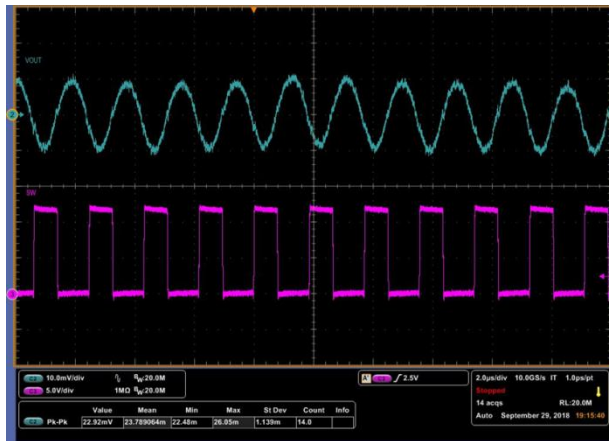
$I_{out}=0.2A$



$I_{out}=0.5A$



$I_{out}=1A$



减小纹波的方法:

- 增大输出电容(若输出为电解电容, 可并联陶瓷电容来减小ESR来减小纹波)
- DCAP: 增大电感可增大轻载开关频率, 从而减小轻载纹波;
PCM: 减小电感可增大轻载开关频率, 从而减小轻载纹波;
- 选择FCCM的芯片(但轻载效率低)

电解电容并联陶瓷电容

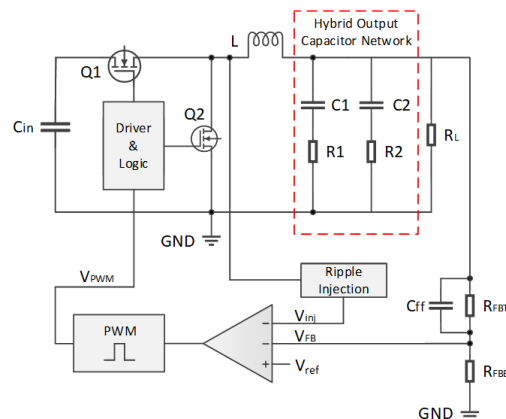
Output Capacitors	Output Voltage Ripple (mV)
22uF (electrolytic)	344
22uF (electrolytic) + 0.1uF (ceramic 0603)	345.5
22uF (electrolytic) + 0.1uF (ceramic 0805)	298.4
22uF (electrolytic) + 0.47uF (ceramic 0805)	209.5
22uF (electrolytic) + 1uF (ceramic 0603)	211.9
22uF (electrolytic) + 1uF (ceramic 0805)	157.3
22uF (electrolytic) + 2.2uF (ceramic 0805)	99.8
22uF (electrolytic) + 4.7uF (ceramic 0805)	56.5
22uF (electrolytic) + 10uF (ceramic 0805)	30.9

Note: 若陶瓷电容容值太小，对减小ESR和输出电压纹波作用不大

输出电容等效ESR:

若 $R_1 \gg R_2$, $C_1 > C_2$, 则

$$Z_{ESR} \approx \frac{R_1}{\left(1 + \frac{C_2}{C_1}\right)^2}$$



关机过程中会重启



原因: 当输入电压跌到UVLO下限值时，芯片停止工作，输入电容不再向负载提供能量，输入电容电压会瞬间增大。若UVLO配置的Vstart和Vstop太接近，输入电容电压超过Vstart, 电源会重启。

解决办法: 调整EN电阻，增大UVLO的迟滞电压。

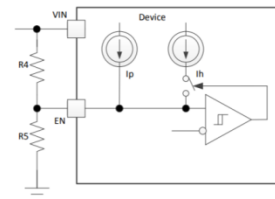


Figure 13. Adjustable VIN Undervoltage Lockout

$$R4 = \frac{V_{START} \left(\frac{V_{ENfalling}}{V_{ENrising}} \right) - V_{STOP}}{I_p \left(1 - \frac{V_{ENfalling}}{V_{ENrising}} \right) + I_h}$$

Where:

$$\begin{aligned} I_p &= 0.7 \mu A \\ I_h &= 1.55 \mu A \\ V_{ENfalling} &= 1.19 V \\ V_{ENrising} &= 1.22 V \end{aligned}$$

$$R5 = \frac{R4 \times V_{ENfalling}}{V_{STOP} - V_{ENfalling} + R4(I_p + I_h)}$$

Q&A?